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Master Thesis Report

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Modeling and Control of Cable-Driven UAV Systems for Collaborative Load Manipulation

Jury

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Abstract

Acknowledgements

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Notations

$F_0(O, x_0, y_0, z_0)$	Inertial reference world frame
$F_p(P, x_p, y_p, z_p)$	Platform reference frame at its center of mass
$F_i(A_i, x_i, y_i, z_i)$	Quadrotor reference frame at cable connection
$F_{C_i}(A_{C_i}, x_{C_i}, y_{C_i}, z_{C_i})$	Quadrotor reference frame attached to the quadrotor at its center of mass
$F_k(A_k, x_k, y_k, z_k)$	Ground vehicle reference frame

Abbreviations

ACT	Aerial Cable Towed
ACTR	Aerial Cable Towed Robot
ACTS	Aerial Cable Towed System
AW	Available Wrench set
CDPR	Cable-Driven Parallel Robot
DKM	Direct Kinematic Model
DoF	Degree of Freedom
IKC	Inverse Kinematic Controller
INDI	Incremental Nonlinear Dynamic Inversion
MPC	Model Predictive Control
NMPC	Nonlinear Model Predictive Control
OCP	Optimal Control Problem
ORCA	Optimal Reciprocal Collision Avoidance
rAW	Radius of the Available Wrench
SAW	Static Available Wrench set
SW	Static workspace
UAV	Unmanned Aerial Vehicle
UGV	Unmanned Ground Vehicle
VIC	Variable Impedance Control
WFW	Wrench Feasible Workspace

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Chapter 1

Introduction

This thesis work takes place in the context of the ShieldBot project, which aims to revolutionize the construction, renovation, and maintenance sectors by investigating and validating a new generation of robotic systems tailored to the industry's unique challenges [1]. The ShieldBot project focuses on three complementary robotic solutions:

- InnerShieldBot is designed for internal construction works to enhance the thermal performance of walls and ceilings.
- InspectionShieldBot is dedicated to identify structural or insulation needs, and ensure long-term building integrity.
- FaçadeShieldBot is designed to operate on building façades and install thermal shielding in order to improve insulation and reduce energy consumption.

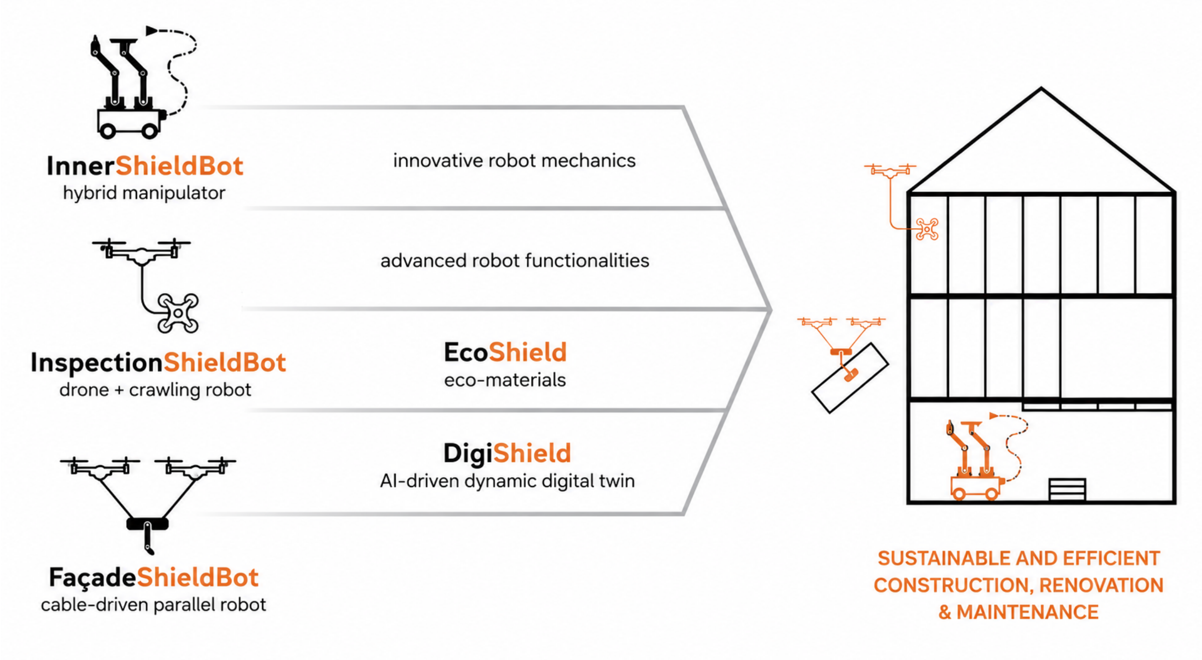

Figure 1.1: Robotic solutions developed by the ShieldBot project [1]

This thesis focuses on the robotic manipulation challenges inherent to the FaçadeShieldBot scenario. The system must generate sufficient force to manipulate the payload while maintaining a stable pose, as the application requires transporting and accurately positioning a payload near a vertical surface. At the same time, it must operate across large façades without requiring complex or permanent infrastructure on the building.

The overall system employs a platform-mounted robotic arm to manipulate and position tools along the surface. However, the scope of this work is strictly centered on the aerial platform itself and its ability to maintain a controlled pose without relying on permanent building infrastructure.

Previous robotic developments have demonstrated the potential of cable-based systems for construction applications. For example, the TitanBot project investigates cable-driven parallel robotic (CDPR) systems capable of transporting loads over large areas and reducing the physical effort required from construction workers [2]. Such systems benefit from the mechanical simplicity and large workspace offered by cable-based actuation. Additionally, their actuators can be located away from the manipulated payload, which makes them particularly attractive for applications involving heavy loads. However, conventional configurations generally rely on fixed or ground-based cable attachment points, which can limit their flexibility when operating over complex or changing environments.

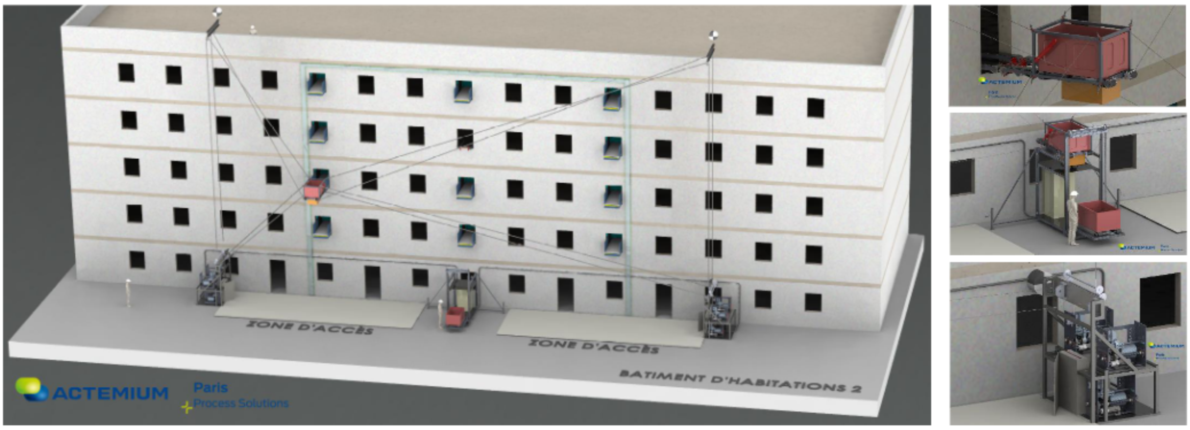

Figure 1.2: Solution proposed by TitanBot project [2]

Aerial robotic systems are a promising approach to access tight and difficult-to-reach areas while reducing the need for traditional scaffolding. Additionally, the use of cables introduces distance between the UAVs and the building, removing direct contact between the two. This is advantageous from a safety and operational perspective, as it reduces the risk of unintended collisions and the impact of the airflow generated by the rotors, allowing the system to safely come into contact with vertical surface [8].

Nonetheless, the position and orientation of the UAVs directly influence the forces and moments that can be generated through the cables. As a result, accurate positioning of the payload can become difficult, particularly when the system is required to perform precise manipulation close to a façade. These limits are especially critical in applications where the payload needs to be precisely positioned and then manipulated against a vertical surface.

The combination of aerial and ground-based actuation can improve the ability of the system to generate and control forces during interaction with the façade while retaining the flexibility of an aerial cable-towed system.

The resulting system, however, is characterized by a large set of design parameters that need to be selected at the design stage and remain fixed during the operation.

They strongly influence the workspace, wrench feasibility, controllability, robustness, and energy efficiency of the system. The main design degrees of freedom include:

- The number of aerial and ground vehicles (UAVs and UGVs), which determines the level of actuation redundancy and force distribution capability;
- The number of cables and their routing, which affect the achievable wrench set and the system's stability;
- The cable length actuation strategy, such as fixed-length or winch-actuated cables;
- The system architecture and graph rigidity, including centralized and decentralized control configurations.

As a result, architectural design cannot be isolated from control system development. A configuration that is mechanically feasible may nevertheless provide poor force distribution or positioning performance in the region of interest. Conversely, an appropriate choice of architecture and attachment locations can improve the controllability and robustness of the system without necessarily increasing its mechanical complexity.

The main objective of this thesis is to study the modelling, design, and control of a hybrid Aerial Cable-Towed System (ACTS) for payload manipulation. In particular, the thesis focuses on how the architecture and cable configuration influence the ability of the system to manipulate a payload and generate the required payload wrench. The work focuses on the selection of suitable cable routing and attachment configurations and on the development of a control approach based on the resulting system model. The proposed configurations are evaluated through simulation to assess their performance and identify the most suitable architecture for the considered application.

The thesis is organized as follows.

Chapter 2

Chapter 3

Chapter 4

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Chapter 6

Chapter 7

Chapter 2
State of the Art

2.1 Cooperative Cable-Driven Aerial Systems

Several robotic architectures can be considered for the manipulation of a payload using multiple aerial vehicles. Among them, Cable-Driven Parallel Robots (CDPRs), Aerial Cable-Towed Systems (ACTSs), and Flying Parallel Robots (FPRs) are particularly relevant to the problem considered in this thesis. These architectures differ mainly in how the payload is connected to the actuators and consequently in their workspace, force-generation capabilities, and mechanical complexity.

2.1.1 Aerial Cable-Towed Systems

An Aerial Cable Towed Robot (ACTR) is a robotic system where the load is connected to multi-rotor UAVs through cables [3]. Other cables may be actuated also by non-aerial devices, such as actuators fixed to the ground or UGVs. In these systems, the payload motion is achieved by coordinating the cable tensions generated by the different actuators.

While standard ACTRs are limited in the directions and magnitudes of forces they can generate, the combination of mobile UAVs and fixed ground anchors enables the system to adapt its available wrench set to task requirements. This allows the platform to exert higher and more controllable interaction forces compared to configurations without a ground-based tension reference [9].

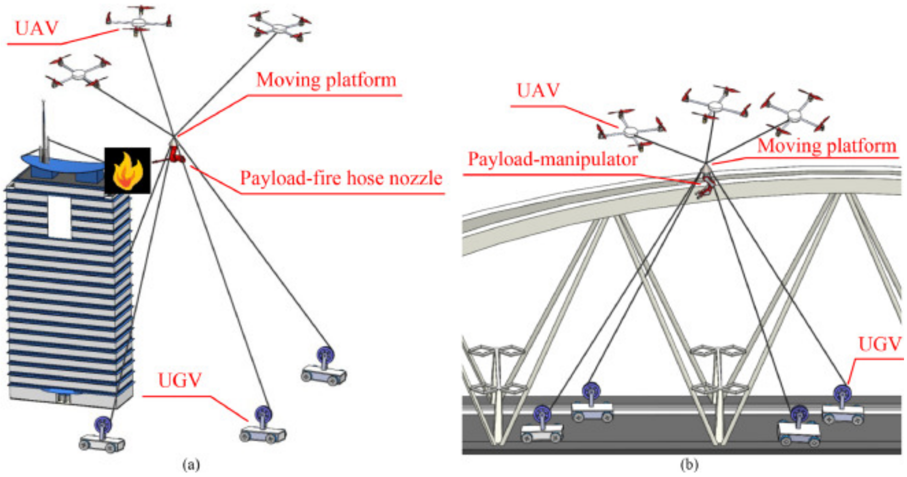


Figure 2.1: ACTRs with land-fixed winches for firefighting applications (a) and repair of a large bridge (b) [3]

2.1.2 Cable-Driven Parallel Robots

In conventional CDPRs, the end-effector is manipulated through flexible cables connected to fixed anchor points on the ground, and its motion is controlled by adjusting the cable lengths. Although CDPRs do not employ aerial robots, their analysis can be beneficial for making progress on ACTSs. As proposed by [8], ACTSs can be interpreted as reconfigurable CDPRs (RCDPRs), where fixed anchor points are replaced by mobile aerial platforms. For this reason, existing results on CDPR kinematics, dynamics, and control should not be ruled out a priori. A fundamental difference between CDPRs and ACTSs lies in the available tension space. In classical CDPR the tension's limits are fixed and do not depend on the robot's pose: the minimum tension is usually a small positive value to avoid slack, while the maximum tension is set by the motor torque. On the other hand, UAVs in ACTSs must always generate enough thrust to keep itself in the air and maintain its attitude, reducing the tension margin available for payload manipulation [10]. Therefore, CDPR findings can still be used for ACTSs, but they need to be adapted by introducing additional constraints due to the quadrotor actuation limits [8, 11].

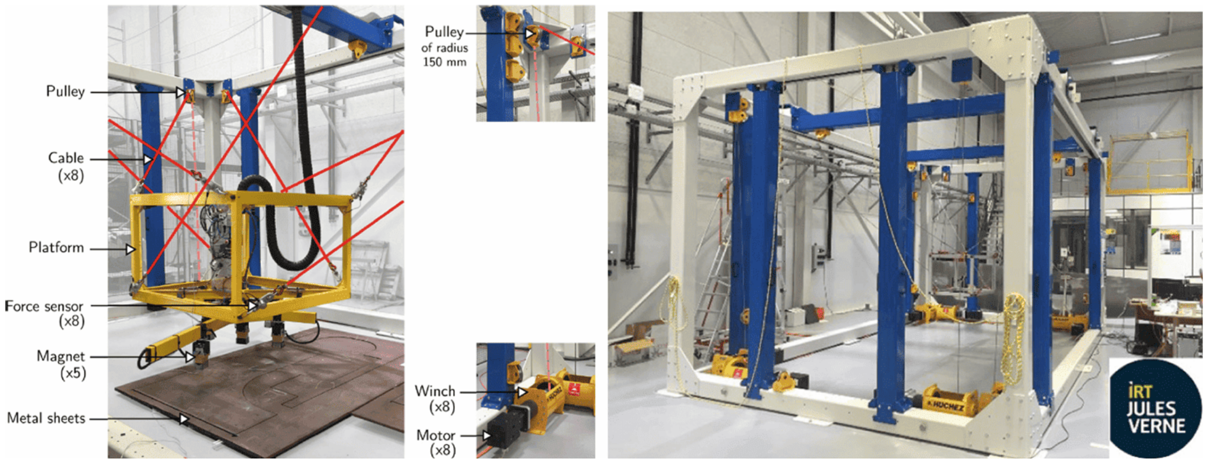


Figure 2.2: CDPR located at IRT Jules Verne [4, 5]

2.1.3 Flying Parallel Robots

Flying Parallel Robots (FPRs) changes from the previous two designs in that they connect the UAVs to the manipulated platform by rigid connections rather than cables, allowing both pushing and pull forces to be transmitted. Thus, the payload can be moved more freely since it has extra passive degrees of freedom inside its legs, also known as kinematic chains. These can be reoriented without actuation, resulting into a platform that isn't over-constrained [12]. While this property can be helpful in interaction tasks, rigid links significantly increases the system's weight and the mechanical complexity [6]. Introducing UGVs in ACTSs as ground anchor points allows pulling forces, expanding the force applicable during interaction tasks. Lastly, even though cables cannot transmit compressive forces, the coordinated control of multiple cable tensions allows the system to counteract reaction forces arising during contact with vertical surfaces, achieving similar objectives without the need for rigid mechanical connections.

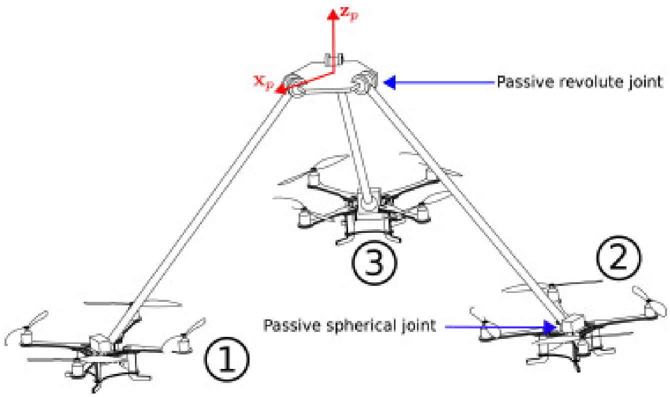


Figure 2.3: Kinematic scheme of the 3-DSR flying parallel robot [6]

2.2 Interaction with the Environment

Physical interaction is particularly challenging for aerial robotic systems because the motion of an aerial vehicle directly affects the forces it generates. Aerodynamic disturbances and modelling uncertainties further complicate the control of the system [12]. In a cable-suspended system, these effects are coupled: a change in the position of one UAV or of the payload modifies the cable directions and consequently the forces experienced by the other agents [13].

These challenges become particularly important when the payload operates close to a vertical surface. Accuracy is a critical requirement in such applications, and interaction with the environment introduces additional force constraints that are not present during free-space motion [9].

Near-façade operation can also be affected by aerodynamic phenomena. Ground and ceiling effects modify the UAV dynamics, while the propeller wash generated by one UAV can disturb other vehicles operating in proximity [6, 13]. Furthermore, uncertainties in payload mass and inertia can affect the required thrust and lead to tracking errors or actuator saturation [8, 13].

Several forms of interaction have been considered for ACT systems, including unexpected collisions, planned contact, and active interaction with an external agent [14]. For the FaçadeShieldBot application, the relevant case is planned physical interaction with a vertical surface.

2.3 Control of Cable-Suspended UAV Systems

2.3.1 Centralized Control

Control architectures for cooperative aerial transportation can broadly be classified as centralized, distributed, or decentralized. In a centralized architecture, information about the UAVs, payload, and system configuration is available to a common controller. This allows the controller to coordinate the agents and explicitly account for the coupling between their motions.

A common approach is to model the ACTS as a continuously reconfigurable cable-driven parallel robot. In this framework, the controller can coordinate the payload motion, cable tensions, and UAV motion within a single control architecture [11, 8].

In [11], the design of a centralized trajectory tracking controller is proposed that explicitly accounts for both the kinematics and dynamics of the system. The control architecture is structured into three main layers:

- A task-space control loop that computes the desired wrench acting on the payload;
- A tension distribution algorithm that determines a feasible set of cable tensions realizing that desired wrench; and
- A joint-space control loop that converts the required cable forces into thrust and attitude commands for each quadrotor.

Differently, in distributed control systems, each agent computes its control action based on locally accessible information, which may be integrated with data resulting from the physical interaction with the shared payload. In this context, consensus algorithms allow individual robots to estimate global properties, such as formation parameters [15].

Lastly, in decentralized approaches, it is crucial that the ACTS maintains bearing rigidity in order to ensure that the system can reconstruct and preserve the formation geometry using only local measurements, avoiding uncontrolled motions and degenerate configurations.

The system considered here is assumed to operate under centralized control, for which the states and configuration of the relevant agents are available to a common controller.

2.3.2 Cable Tension and Force Control

Cable tension management is fundamental to the control of cable-driven systems because cables can only transmit tensile forces. The controller must therefore ensure that the tensions remain positive to prevent cable slackness and maintain the desired mechanical configuration.

In redundant cable-driven systems, several sets of cable tensions can generate the same net wrench on the payload. This redundancy can be exploited to satisfy additional constraints while maintaining the desired payload wrench. In particular, internal forces corresponding to the null space of the wrench mapping can be used to modify the tension distribution without changing the resulting payload wrench [11, 9].

For ACTSs, tension feasibility is further constrained by the UAVs' actuation limits. Each UAV must allocate part of its available thrust to compensate for its own weight and maintain its attitude, leaving only a limited portion of its actuation capability for payload manipulation [10]. A valid control solution must therefore consider both the desired payload wrench and the available actuation capabilities.

More recent approaches incorporate cable tension constraints directly into trajectory optimization, allowing future trajectories to be evaluated according to their feasibility with respect to cable tension and actuator limits [13]. Such approaches demonstrate the importance of considering cable forces explicitly when planning or controlling cooperative aerial systems.

CHAPTER 3

ACTS Design and Modeling

3.1 System Architecture

To formulate a general methodology, it is important to adopt a parameterization of the ACTS. We consider an ACTS with n_q quadrotors, n_g ground vehicles, m cables, and a load with d degrees of freedom. The i-th quadrotor is connected to the load at point B_i through a cable of length l_i, whose other end is fixed to the vehicle at point A_i, for $i = 1, \dots, n_q$. Similarly, the j-th ground vehicle is connected to the load at point B_j via a cable of length l_j, attached to the vehicle at point A_j, for $j = 0, \dots, n_g$. To avoid ambiguity between aerial and ground actuators, alphabetical indices may be used in the latter case to distinguish the two sets of vehicles [10].
We can also define the following systems of reference:

- $F_0(O, x_0, y_0, z_0)$ as the inertial reference world frame;
- $F_p(P, x_p, y_p, z_p)$ as the frame attached to the platform at its center of mass;
- $F_i(A_i, x_i, y_i, z_i)$ as the frame attached to the quadrotor at point A_i, with the z-axis normal to the plane of the rotors;
- $F_{C_i}(A_{C_i}, x_{C_i}, y_{C_i}, z_{C_i})$ as the frame attached to the quadrotor at its center of mass A_{C_i} with the x-axis pointing forward, the y-axis pointing left and the z-axis pointing upwards;
- $F_k(A_k, x_k, y_k, z_k)$ as the frame attached to the ground vehicle at point A_k, with the z-axis normal to the ground.

In particular, since the ground vehicles are considered fixed in pose, also their reference frame is inertial and the fixed in time transformation matrix T_k from the reference world frame to each ground vehicle is known.

In this thesis, the following assumptions are adopted:

- The cables are modeled as massless, non-extensible and perfectly straight [14, 10, 8];
- The cable attachment point coincides with the quadrotor's center of mass [6, 10, 12];
- There are no aerodynamic disturbances, such as the rotor wash [6, 12, 15];
- The flight is quasi-stationary and the aerodynamic friction and resistance can be neglected in the dynamic model [16, 10];

Lastly, the ground vehicles will be considered fixed anchoring points, as the ShieldBot project is still at its early stages.

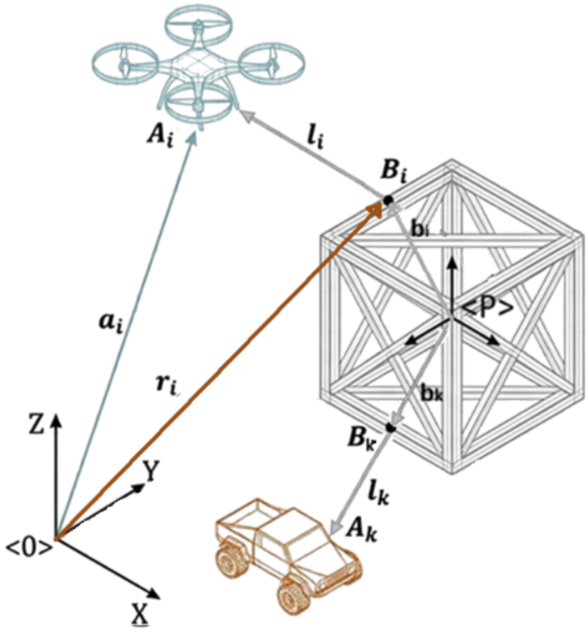


Figure 3.1: Graphical representation of the platform

3.2 Kinematostatic model

As talked about in Section 2.1.2, it is possible to model cable-suspended UAV systems as reconfigurable cable-driven parallel robots (RCDPRs), where the traditional fixed actuators are replaced by mobile UAVs, possibly integrated with land-fixed winches to balance interaction forces at high altitudes.

A kinematostatic model is a model that combines kinematics (motion and geometry) with statics (forces and equilibrium) to describe how a mechanical system moves and how forces are transmitted through it.

3.2.1 Kinematic model

Given the payload twist ${}^0\xi = \begin{bmatrix} {}^0v_p & {}^0\omega_p \end{bmatrix}^\top$ in the inertial reference frame $< 0 >$, we can write the position and velocity of the attachment point B_i (resp. B_k) of the aerial vehicle (resp. ground) in $< 0 >$.

$$ {}^0\boldsymbol{r}_i = {}^0 \boldsymbol{p} + {}^0_P R \, {}^P\boldsymbol{b}_i \tag{3.1} $$

where b_i is the attachment point on payload.

$$ {}^0\dot{\boldsymbol{r}}_i = {}^0 \boldsymbol{v}_p + {}^0 \boldsymbol{\omega}_p \times \left({}^0_P R \, {}^P\boldsymbol{b}_i \right) \tag{3.2} $$

For definition, the cable vector is $\boldsymbol{l}_i = \boldsymbol{a}_i - \boldsymbol{r}_i = A_i - B_i$ with $\boldsymbol{a}_i$ being the anchor position. Under a centralized controller, the positions of all UAV A_i and UGV A_k are known which directly determine the anchor vector $\boldsymbol{a}_i$. Furthermore, the payload's position and orientation are assumed to be known together with the known locations of the attachment points relative to the payload, this information allows us to calculate the cable vectors $\boldsymbol{l}_i$.

The final form of cable vector variation is given by applying Equation 3.2 in its definition:

$$\dot{\boldsymbol{l}}_i = \dot{\boldsymbol{a}}_i - \dot{\boldsymbol{r}}_i = \dot{\boldsymbol{a}}_i - \frac{d}{dt}\left(\boldsymbol{p} + {}_P R\, {}^P\boldsymbol{b}_i\right) = \dot{\boldsymbol{a}}_i - \left(\boldsymbol{v}_p + \boldsymbol{\omega}_p \times \left({}_P R\, {}^P\boldsymbol{b}_i\right)\right) \tag{3.3}$$

Each $\dot{l}_i$ corresponds to the projection of the relative velocity of B_i along the cable direction

$$\begin{aligned} \dot{l}_i &= u_i^\top \left(\dot{\boldsymbol{a}}_i - \left(\boldsymbol{v}_p + \boldsymbol{\omega}_p \times \left({}_P R\, {}^P\boldsymbol{b}_i\right)\right)\right) \\ &= u_i^\top \dot{\boldsymbol{a}}_i - \left[u_i^\top \quad \left({}_P R\, {}^P b_i \times u_i\right)^\top\right] \xi \\ &= J_{a,i}\, \dot{a}_i - J_{p,i}\, \xi, \end{aligned} \tag{3.4}$$

This gives us the kinematic model that relates the variation of cable length vector $\boldsymbol{\rho}$ with the change in position of the anchors and the payload twist:

$$\dot{\rho} = J_a \dot{a} - J_p \xi = J_a \begin{bmatrix} \dot{a}^{(a)} \\ \dot{a}^{(g)} \end{bmatrix} - J_p \xi \tag{3.5}$$

Let's $m = n_a + n_g$ be the total number of winches (or cables) and $u_i = l_i / \|l_i\|$ the unit cable direction, then:

- $\dot{\rho} = \begin{bmatrix} \dot{l}_1 & \dot{l}_2 & \dots & \dot{l}_m \end{bmatrix}^\top$ is the cable length rate (dimension $[m \times 1]$).
- J_a is the Jacobian relating the anchor velocities to the cable-length rates $[m \times 3m]$. In particular, each cable depends only on its anchor velocity.

$$J_a = \begin{bmatrix} u_1 & 0 & \cdots & 0 \\ 0 & u_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & u_{n_a+n_g} \end{bmatrix}, \qquad u_j = \frac{(a_j - r_j)}{\|a_j - r_j\|}. \tag{3.6}$$

- J_p is the Jacobian relating the payload twist to the cable-length $[m \times 6]$

$$J_p = \begin{bmatrix} u_1^\top & \left({}_p R\, {}^p b_1 \times u_1\right)^\top \\ \vdots & \vdots \\ u_m^\top & \left({}_p R\, {}^p b_m \times u_m\right)^\top \end{bmatrix} \tag{3.7}$$

The control allocation matrix maps the actuator inputs (cable tension vector) to the system (payload) wrench. For this definition $J_p^\top$ is the allocation matrix.

$$W_p = J_p^\top \tau \tag{3.8}$$

Let's W_p^* being the desired payload wrench, the control allocation problem to solve is

$$J_p^\top \tau = W_p^*, \qquad \text{s.t.} \tau_i > 0 \tag{3.9}$$

3.2.2 Static model

Let $\tau = [\tau_{uav}^\top, \tau_{ugv}^\top]^\top \in \mathbb{R}^{m \times 1}$ the cable tension vector, then

$$\begin{aligned} \underbrace{\tau^\top \dot{\rho}}_{\text{total cable power}} &= \underbrace{\tau^\top J_a \dot{a}}_{\text{UAVs and UGVs mechanical power}} - \underbrace{\tau^\top J_p \xi}_{\text{payload power}} \\ &= (J_a^\top \tau)^\top \dot{a} - (J_p^\top \tau)^\top \xi \\ &= F_a^\top \dot{a} - W_p^\top \xi \end{aligned} \tag{3.10}$$

Each cable i applies a force on the payload $F_{a,i}$ and an equal and opposite force on the UAV (resp. UGV). We can write balance equation on our elements:

Payload Balance

$$\underbrace{W_p}_{J_p^\top \tau} + \underbrace{W_g}_{\begin{bmatrix} 0 \\ 0 \\ -m_p g \\ 0 \\ 0 \\ 0 \end{bmatrix}} + W_{contact} + \underbrace{W_{wind}}_{\text{let } = 0} = 0 \tag{3.11}$$

where since each cable produces a wrench on the payload, the contribution given by the i-th cable is:

$$\underbrace{w_{p,i}}_{6\times 1} = \underbrace{J_{p,i}^\top}_{6\times 1} \underbrace{\tau_i}_{1\times 1} = \begin{bmatrix} u_i^\top & ({}_P R\, {}^P b_i \times u_i)^\top \end{bmatrix}^\top \tau_i = \begin{bmatrix} \tau_i u_i \\ ({}_P R\, {}^P b_i \times \tau_i u_i) \end{bmatrix} \tag{3.12}$$

and $W_{contact} = 0$ when there is no contact with the environment.

UAV balance

$$-F_{a,i} + F_{prop,i} + \underbrace{F_{g,i}}_{\begin{bmatrix} 0 \\ 0 \\ -m_i g \end{bmatrix}} = 0 \tag{3.13}$$

with ${}^i F_{prop} = \begin{bmatrix} 0 & 0 & F_{prop,z,i} \end{bmatrix}^\top$ and $F_{a,i} = J_{a,i}^\top \tau_i = \tau_i u_i$.

UAVs must always generate sufficient thrust to maintain flight and ensure their own stability. As a result, only a portion of their available thrust can be allocated to payload manipulation, which reduces the effective force that can be transmitted through the cables.

UGV balance

$$-F_{a,j} + \underbrace{F_{ground,j}}_{\begin{bmatrix} F_{frict,x,j} \\ F_{frict,y,j} \\ F_{norm} \end{bmatrix}} + \underbrace{F_{g,j}}_{\begin{bmatrix} 0 \\ 0 \\ -m_j g \end{bmatrix}} = 0 \tag{3.14}$$

with $\|F_{frict}\| \leq \mu F_{norm}$, $F_{norm} = m_j g - F_{a,z,j}$ and $F_{a,j}$ is obtained through Equation 3.15.

Similarly, UGVs are subject to ground interaction constraints. In particular, they must satisfy friction limits to remain stationary when no motion is commanded. This imposes bounds on the admissible cable forces, as excessive tension may cause slipping. By Newton's third law, the force the cable exerts on the drone $F_{a,i}$ (resp. on the UGV $F_{a,j}$) is equal and opposite to the force it exerts on the payload. Since u_i points from B_i toward A_i, the cable pulls the drone with

$$F_{a,i} = J_{a,i}^\top \tau_i = \tau_i u_i \tag{3.15}$$

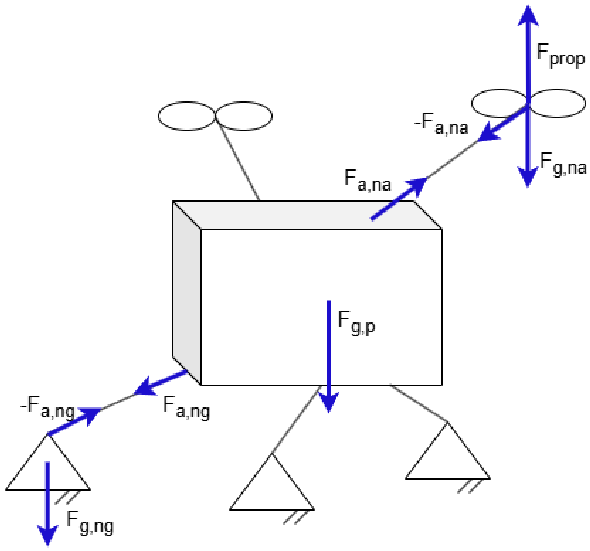


Figure 3.2: Forces acting on the ACTS

As a consequence, the total cable force on the payload is:

$$F_p = -\sum_{i=1}^{n_a} F_{a,i} - \sum_{j=n_a+1}^{n_a+n_g} F_{a,j} \tag{3.16}$$

where the payload wrench is $W_p = \begin{bmatrix} F_p & M_P \end{bmatrix}^\top$

3.2.3 Kinematostatic model

Writing Equation 3.11, 3.13 and 3.14 in terms of τ, we obtain:

$$W_{sys} = \begin{bmatrix} W_p \\ F_a \end{bmatrix} = \begin{bmatrix} J_p^\top \\ J_a^\top \end{bmatrix} \tau = G\tau = \begin{bmatrix} -W_g - W_{contact} - W_{wind} \\ F_{prop} + F_{ground} + F_g \end{bmatrix} \tag{3.17}$$

where $F_{prop,j} = 0$ for the UGVs and $F_{ground,i} = 0$ for the UAVs. Then W_{sys} is $(6+3m) \times 1$ and

$$G\tau + \begin{bmatrix} W_g + W_{contact} + W_{wind} \\ -F_{prop} - F_{ground} - F_g \end{bmatrix} = 0 \tag{3.18}$$

3.3 Architecture Design

The configuration of a cooperative aerial transportation system (ACTS) has a direct influence on its workspace, force-generation capabilities, controllability, and energy consumption. For a given payload pose, multiple equilibrium configurations may exist, corresponding to different distributions of internal cable forces that do not modify the resulting net wrench acting on the payload. Although these configurations satisfy the same static equilibrium conditions, they can require significantly different UAV positions and cable tensions, leading to different energy consumption and robustness properties [17]. Consequently, the architecture of the ACTS must be selected according to the requirements of the intended application.

3.3.1 Design Variables

Each configuration is characterized by parameters such as the number and arrangement of aerial and ground actuator, the cable attachment points, and the cable routing. More specifically, the ACTS consists of at least one UAV, required as the payload needs to be lifted from the ground, and one UGV, included for safety and operational constraints. Lastly, for a rigid payload with six degrees of freedom, at least six independent cable are required to generate a general six-dimensional payload wrench.
In this thesis, we suppose working with nine actuators, three aerial and six on the ground. Since we are interested in working with an over-actuated system, in this thesis we suppose working with nine actuators, three aerial and six on the ground. While the latter are used to fully control the payload pose, the three drones will act as an anti-gravitational force. A schematic is shown in Figure 3.3. The candidate architectures are subsequently evaluated using the performance indices introduced in Section 3.3.2.

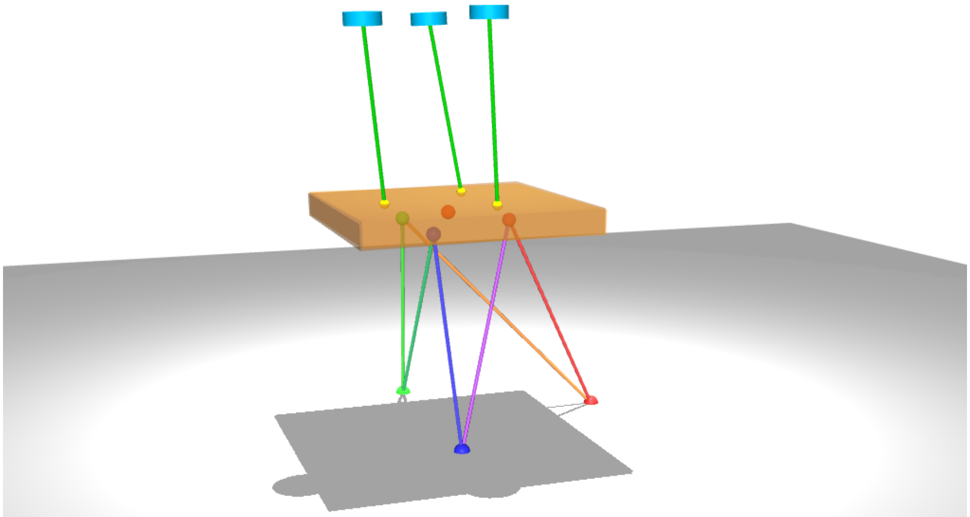


Figure 3.3: Over actuated ACTS with 6 ground cables and 3 UAVs

Aerial Actuators

The configuration of the UAV connections is crucial, as it directly affects how forces are applied to the payload: UAVs are responsible for generating the lifting force required to lift the payload. Consequently, their number depends strongly on the payload mass. However, an increased number of UAVs does not necessarily improve the performance of the system, as this increase can lead to higher power consumption and greater system complexity. Additionally, each UAV can be connected to the payload using either a single cable or multiple cables. The choice of configuration affects controllability, wrench feasibility, robustness, and energy efficiency and should be evaluated accordingly. A practical compromise is to use three UAVs connected to the payload at distinct attachment points, providing a balance between the metrics just mentioned (Figure 3.4).

As we will see in later discussions, the main alternatives considered in this work are illustrated in Figure 6.3.

Ground Actuators and Anchors

In contrast, UGVs do not lift the payload, so the same stability arguments are not valid. If we treat the UGVs as fixed anchor points, connecting a single UGV to multiple hook points on the payload simplifies to a standard fixed-anchor scenario. Because each actuator operates independently, this configuration adds no extra mathematical complexity. However, if the UGVs are actively moving, multi-cable connections require precise winch coordination to ensure all cables remain taut at all times. In Figure 3.5, possible payload hook attachments can be found.

The number and spatial distribution of these anchors affect the directions of the corresponding cable forces and therefore the wrench that can be generated at the payload. Different anchor layouts can consequently lead to different force and moment capabilities.

Payload Attachment Points

The locations of the cable attachment points on the payload directly affect the moments generated by cable tensions. Attachment points that are more widely separated can provide larger moment arms, whereas closely spaced attachment points result in different wrench and conditioning properties.

The attachment points are therefore treated as design variables in the architecture selection process.

Ground Anchor Placement and Cable Routing

The positions of the ground anchors determine the directions of the ground cables for a given payload pose. Their arrangement consequently affects both the achievable wrench and the distribution of cable tensions.

In addition to anchor placement, the routing of the cables between the ground anchors and the payload attachment points is considered as a design variable. Candidate routings include crossed and uncrossed arrangements, as well as adjacent and diagonal connections. These different routings can produce different cable directions and therefore different force-generation capabilities.

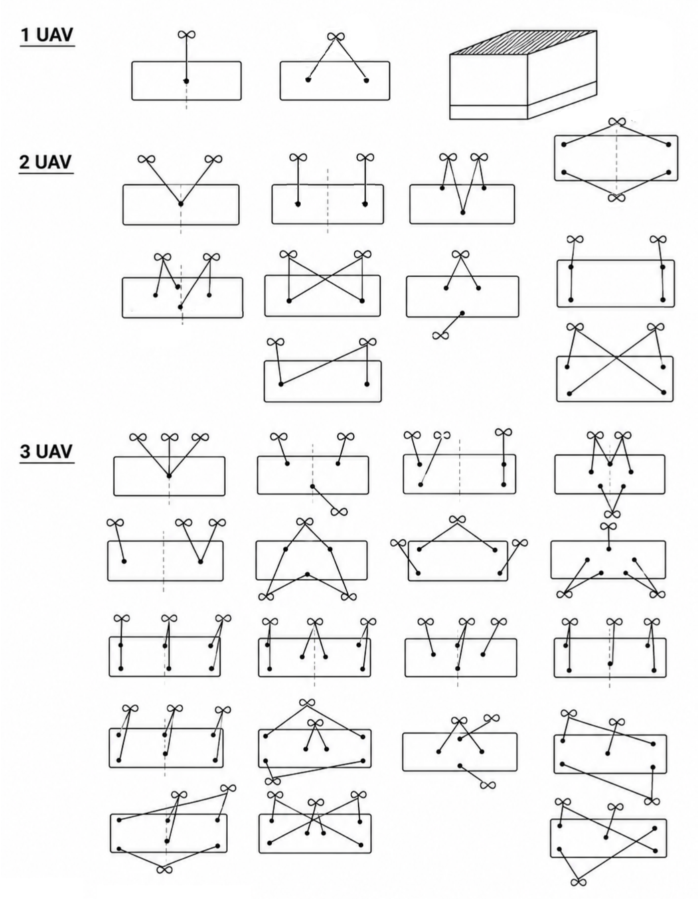


Figure 3.4: Possible UAVs connection to the payload

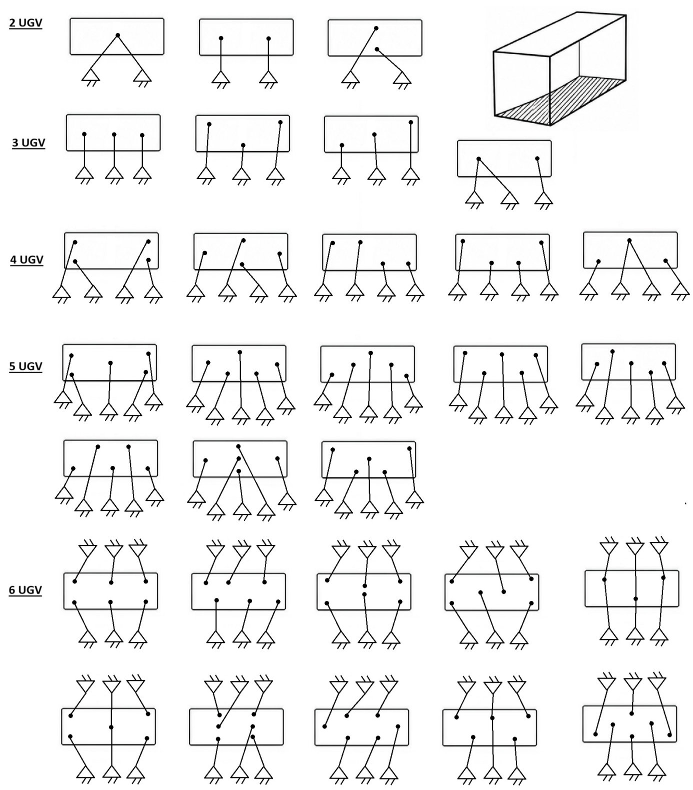


Figure 3.5: Possible UGVs connection to the payload

3.3.2 Performance Metrics

In the context of the ShieldBot project, we are working with two questions:

1. Which attachment geometry gives the best ability to generate the required payload wrench over the entire mission workspace?
2. Given the chosen architecture, where should the actuators (UAVs and anchor points in the ground) be positioned, and what cable lengths should be used during operation?

In both cases, we need an objective measurement that guides us in the selection.
Every configuration to be eligible needs to satisfy feasibility metrics first, such as [18]:

- Wrench closure conditions (WCC), which guarantees that the payload is fully constrained and arbitrary wrenches can be generated. Mathematically: $\text{rank}(J_p) = 6$ and positive tensions exist.
- Wrench Feasible Condition (WFC), which checks whether tensions remain in the feasible bound $0 \leq \tau_{\min} \leq \tau \leq \tau_{\max}$, included UAV thrust limits, cable strength limits and UGV friction limits in a future where the ground anchor point will be movable.
- Interference-Free Condition (IFC), ensuring that there is no cable rubbing and no collision with payload edges. Mathematically:

$$\begin{aligned} d_{ij} &\geq d_{\text{safe}}, \quad \text{where } d_{ij} = \text{dist}\,(\text{cable } i, \text{cable } j) && \forall i, j \\ d_i &\geq d_{safe}, \quad \text{where } d_i = \min_{s\in[0,1]} \left(\text{dist}\,(r_i + s(a_i - r_i), P)\right) && \forall i \end{aligned} \tag{3.19}$$

  where P is the payload geometry and d_{safe} is a clearance threshold.

However, assuming that all cable attachment points are on the same face of the payload, the only feasible interference between the latter and one of the cables is a near-tangent cable that touches that face.

After passing the feasibility filters, we need a way to assign a score to each hook configuration.

Capacity margin or minimum degree of constraint satisfaction

The capacity margin is a robustness index that is used to analyze the degree of feasibility of a configuration. The feasibility region is the set of all poses and geometric configurations in which the system can physically balance a required task wrench, such as gravity or external interaction forces, while satisfying the quadrotor's actuation limits and the physical constraints of the cables [10]. It is defined as the shortest signed distance from the task wrench to the boundary of the available wrench set (AWS). Mathematically, it is the minimum of the distances of each one of the n vertices $w_{e,j}$ to each one of the p facets $(\mathbf{a}_l, b_l)$ of the AWS.

$$\gamma = \min_{j=1\div n} \min_{l=1\div p} \gamma_{j,l}, \quad \text{where } \gamma_{j,l} = \frac{b_l - w_{e,j}^\top \mathbf{a}_l}{\|\mathbf{a}_l\|_2} \tag{3.20}$$

where $\mathbf{a}_l$ is the normal vector to the l-th facet and b_l is its offset [19].

The wrench vector combines force components (N) and moment components (Nm), which are not directly comparable in a Euclidean sense. Computing the capacity margin γ by measuring facet distances in the raw, mixed-unit wrench space would make γ depend on the arbitrary choice of length unit used to express the cable attachment points, rather than on the physical robustness of the configuration. Following Ruiz et al. [19], the moment components of the wrench are therefore rescaled by a characteristic length.

$$r = \sqrt{\frac{1}{m}\sum_i \|b_i\|^2} \tag{3.21}$$

Where b_i is the position of the i-th cable attachment point relative to the payload center of mass, before the capacity margin is evaluated. This rescaling renders all six wrench components dimensionally homogeneous (in force units) and ensures γ is independent of the chosen length scale.

Worst-Case Capacity Margin

Additionally, we can define the worst-case capacity margin as to how far the system remains from actuator saturation in the most demanding operating condition. Mathematically:

$$\zeta = \min\left(1 - \max_i \frac{\tau_i}{\tau_{i,max}}\right) \tag{3.22}$$

Radius of Available Wrench (rAW)

The radius of the available wrench (rAW) is the maximum magnitude of force that the system can generate uniformly in all directions while remaining within admissible tension limits [9]. Its optimization helps identify arrangements that simultaneously maximize force robustness, geometric and collision-avoidance constraints, resulting into an enlargement of the static workspace [9].

$$rAW = \min_i\left(\min\left(\frac{|d_{pi}|}{\|\mathbf{n}_{null,i}\|}, \frac{|d_{qi}|}{\|\mathbf{n}_{null,i}\|}\right)\right) \tag{3.23}$$

being $u_{null,i} = \text{null}(M_i)$ with M_i containing combinations of linearly independent cable unit vectors, d_{pi} and d_{qi} defined as:

$$\begin{aligned} d_{pi} &= \sum_{j\in I^+} \tau_{max} u_{nulli}^T u_j + \sum_{j\in I^-} \tau_{min} u_{nulli}^T u_j - m|u_{nulli}^T g| \\ d_{qi} &= -\sum_{j\in I^-} \tau_{max} u_{nulli}^T u_j - \sum_{j\in I^+} \tau_{min} u_{nulli}^T u_j - m|u_{nulli}^T g| \end{aligned} \tag{3.24}$$

Since the formulation of rAW in [9] considers a point-mass payload, it does not account for the moments generated by the cable tensions about the payload's attachment points. To capture both effects, the radius of the available wrench is therefore split into two independent quantities: rAW$_{force}$, the maximum pure force the system can generate uniformly in any direction while producing zero net moment, and rAW$_{moment}$, the analogous quantity for pure moment generation with zero net force. Whereas rAW$_{force}$ admits the closed-form null-space expression of Eq. 3.23, rAW$_{moment}$ is instead obtained through a vertex-enumeration procedure: candidate cable tension vectors satisfying zero net force at the bounds of the admissible tension box are computed, their corresponding moments $\tau_i(\mathbf{b}_i \times \mathbf{u}_i)$ are collected, and rAW_{moment} is taken as the distance from the origin to the nearest facet of the convex hull of these moment vectors.

Conditioning index

The conditioning index measures how uniformly the system can generate forces and moments in different directions. Mathematically:

$$\nu = \frac{\sigma_{\min}(J_p)}{\sigma_{\max}(J_p)} \in [0,1] \tag{3.25}$$

$\sigma_{\min}$ and $\sigma_{\max}$ are the minimum and maximum singular values of the Jacobian matrix, obtained via Singular Value Decomposition (SVD) [15]. A small conditioning number indicates a well-conditioned system, far-away from singular configuration.

Manipulability

The manipulability index, defined as $w = \sqrt{\det(J_p J_p^\top)}$, measures the volume of the wrench ellipsoid. A value close to 0 means that the system hit a kinematic singularity, while large value means a large overall wrench generation capability.

Composite score

Once the feasibility constraints are satisfied, we may want to proceed either by

$$N = w_1 \hat{\gamma} + w_2\, \mathrm{r\hat{A}W} + w_3 \hat{\zeta} \tag{3.26}$$

where all the metrics are scaled to $[0,1]$,

$$U^* = \arg\max_U \left(N(U)\right) \tag{3.27}$$

3.3.3 Selection Strategy

The architecture search involves a potentially large number of candidate ground-cable routings. Evaluating the complete nonlinear ACTS model for every candidate would be computationally expensive, particularly because the UAV positions and cable directions depend on the selected architecture. Therefore, a two-tier selection strategy is adopted to progressively eliminate unsuitable configurations before performing the more expensive full-system analysis.

Initially, candidate ground-cable routings are screened using the ground-cable subsystem alone. For each routing, the ground-only payload Jacobian ($J_{p,\text{ground}}$) is constructed from the six ground cables and evaluated against the Wrench Closure Condition (WCC), $\text{rank}(J_{p,\text{ground}}) = 6$, and the Interference-Free Condition (IFC), restricted to ground-cable pairs. A routing is discarded immediately if it violates either criterion at any tested pose.

The surviving ground routings are subsequently combined with candidate UAV attachment geometries. For each resulting complete ACTS configuration, the full payload Jacobian, $J_{p,\text{full}}$, is assembled from the ground and aerial cables. Wrench feasibility is then evaluated by solving the bounded optimization problem

$$\min_{\boldsymbol{\tau}} \left\| J_{p,\text{full}} \boldsymbol{\tau} - \mathbf{W}^* \right\|^2, \qquad \text{subject to} \qquad \boldsymbol{\tau}^*_{\min} \leq \boldsymbol{\tau} \leq \boldsymbol{\tau}^*_{\max}, \tag{3.28}$$

where $\mathbf{W}^*$ denotes the desired payload wrench and the tension bounds account for the admissible cable tensions, including the maximum tension imposed by the UAV thrust capabilities.

A configuration is considered wrench-feasible when the minimum residual of Eq. 3.28 is below a prescribed tolerance. The surviving configurations are subsequently compared using the performance metrics defined in Section 3.3.2.

This strategy separates the inexpensive combinatorial screening of ground-cable routings from the more computationally demanding analysis of the complete aerial-ground system. It therefore reduces the number of configurations requiring full nonlinear evaluation while retaining the relevant UAV-dependent constraints for the final architecture selection.

Chapter 4

Control of the ACTS

4.1 Force control with a point-mass payload

Before going into details on the full consideration, let's analyze some simpler configurations.

4.1.1 Case: one drone connected to the ground through a cable

The goal is to design the drone force F_{prop} such as that the ground tension is controlled. Since only a tension in the cable direction is imposable, an arbitrary force f^* at the ground anchor can be achieved only by moving the drone, changing the cable direction u. Called a the drone's position and b the attachment point to the ground:

$$u = \frac{a - b}{\|a - b\|} \tag{4.1}$$

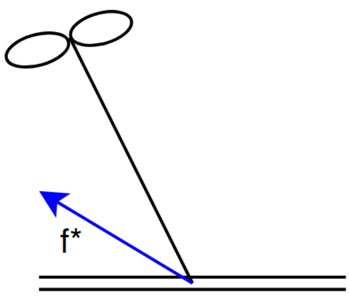


Figure 4.1: One drone system connected to the ground through a cable

Option 1

Defined the desired cable tension τ^* and the desired cable direction u^* as:

$$\tau^* = \|f^*\| \tag{4.2}$$

$$u^* = \frac{f^*}{\|f^*\|} \tag{4.3}$$

The ground tension is obtained using the geometric constraint which relates the anchor point to the drone position:

$$a^* = b + l\, u^* \tag{4.4}$$

where l is the cable length. Equation 4.4 is then used as a reference to compute the propeller force.

$$F_{prop} = m_i g e_3 + K_p(a^* - a) + K_d(\dot{a}^* - \dot{a}) + \tau^* u \tag{4.5}$$

Option 2

Given the drone Newton's Second Law described in Equation 3.13, we are able to estimate the cable force $\hat{f} = F_{a,i}^{est}$

$$m_i \ddot{a}_i = F_{prop,i} - \hat{f} + F_{g,i} \qquad \rightarrow \qquad \hat{f} = F_{prop,i} + F_{g,i} - m_i \ddot{a}_i \tag{4.6}$$

where all the forces are in the world frame and $\ddot{a}_i$ is the drone linear acceleration. Using Equation 3.13, we would like to apply a propulsion force equal to

$$F_{prop,i}^* = f^* - F_{g,i} \tag{4.7}$$

Finally, the closed loop equation is:

$$F_{prop,i} = F_{prop,i}^* + K_p e + K_d \dot{e}, \quad \text{with } e = f^* - \hat{f} \tag{4.8}$$

The result is given to the attitude controller, which aligns the body z-axis with the commanded thrust vector.

4.1.2 Case: payload connected to the ground and a drone

In this case, we are working with the simplified case of an ACTS composed by a drone, a ground anchor and a point-mass payload. Called with subscript 1 the cable that connects the payload to the drone and with 2 the one that connects it to the ground, for definition the vector u_i, $i = 1 \div n$ is given by

$$u_i = \frac{a_i - p}{\|a_i - p\|} \tag{4.9}$$

being a_i and p respectively the actuator and the payload position.

Payload position control

The goal is to design $F_{prop,1}$ such that the payload reaches a desired position ($p \rightarrow p^*$), which is reachable for the length of the cable.

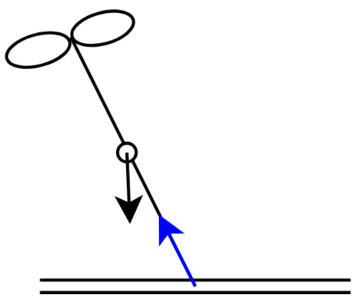


Figure 4.2: Simplified ACTS with one drone and one ground anchor

The payload dynamics is given by:

$$\begin{aligned} m_p\ddot{p} &= \tau_1 u_1 + \tau_2 u_2 - m_p g e_3 \\ &= F_{a,1} + F_{a,2} + F_g \qquad \rightarrow \qquad F_p = m_p(\ddot{p} + g e_3) \\ &= F_p + F_g \end{aligned} \tag{4.10}$$

The desired payload force is defined using an impedance controller:

$$F_p^* = m_p(\ddot{p}^* + g e_3) + K_p(p^* - p) + K_d(\dot{p}^* - \dot{p}) \tag{4.11}$$

with $\dot{p}^* = 0$ and $\ddot{p} = 0$ as we required the payload to maintain the position once it is reached. The remaining force to be generated by cable 1 is:

$$F_{a,1}^* = F_p^* - F_{a,2} \tag{4.12}$$

where $F_{a,2} = \tau_2 u_2$. This brings us back to the situation in Section 4.1.1, where $F_{a,1}$ is f^*.

Tension and orientation control

The goal can also be design $F_{prop,1}$ such that an arbitrary force f^* on the ground can be applied ($F_2 \rightarrow f^*$). In other words, $u_2 \rightarrow u_2^*$ and $\tau_2 \rightarrow \tau_2^*$. This case can be interesting to study as in a redundancy system situation, it may be helpful to add additional constraint in one of the cables. The only difference from the previous case is that the desired position for the payload p^* is not an input anymore, but obtained such that the 2nd cable is aligned along the desired direction given by f^* (Equation 4.2 and 4.3). Then:

$$p^* = a_2 - l_2 u_2^* \tag{4.13}$$

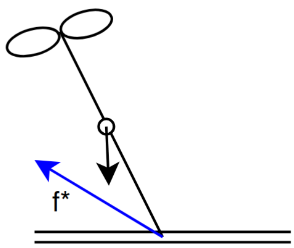


Figure 4.3: Tension directed arbitrary

4.1.3 Case: payload connected to the ground through two cables

The ACTS consists of a point-mass payload connected to the ground through two cables and to an UAV. The enumeration is shown in Figure 4.4.

Payload position control

If the input is the desired payload position p^*, we find ourselves in three situations:

1. p^* lies precisely on the intersecting boundary arcs dictated by the maximum lengths of the ground cables;

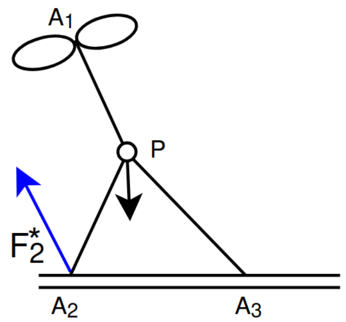


Figure 4.4: Payload connected through two cable to the ground

2. p^* is out of reach;

3. p^* is inside the triangle A_1A_3P, resulting into slack cables

In the first case, p^* is used to compute the desired payload force F_p^* (Equation 4.11). Using the simplified payload balance equation related to the forces acting on the payload (Equation 3.16), we obtain the following.

$$F_{a,1}^* = F_p^* - \sum_{i=2}^{n} F_{a,i} \tag{4.14}$$

which brings us back to the previous case.

Because the payload is physically constrained at a boundary where both ground cables are stretched straight, any movement or external disturbance is perfectly resisted by the tension of the cables.

In the second case, since the ground cables physically block the payload from reaching p^*, a steady error $e = (p^* - p) \neq 0$ builds up. The actual tension will be directly proportional to the control gains and the distance of the unreachable input:

$$\sum F_{virtual} \propto (p^* - p_{geom}) \quad \Longrightarrow \quad \sum F_{virtual} = K(p^* - p_{geom}) \tag{4.15}$$

where p_{geom} is the closest reachable point that lies on the workspace boundary. In this case, we may impose a controlled tension on the ground cables directed along $(p^* - p_{geom})$ to ensure the drone continues to pull toward the desired target:

$$F_{a,1}^* = F_p^* - \sum_{i=2}^{n} F_{a,i} + F_{virtual} \tag{4.16}$$

In the last case, the distance from the anchors to the payload is less than the cable length, resulting into slack cables as the latter cannot push. Mathematically $\tau_2 = 0$ and/or $\tau_3 = 0$. Then, $F_{a,1}^* = F_p^*$

Desired tension in the cable on the ground

The desired variable to control is the tension of one of the cables $F_{a,2} \to f^*$.

$$F_{a,1}^* = F_p - f^* - F_{a,2} \tag{4.17}$$

where $F_p = m_p(\ddot{p} + ge_3)$ and $F_{a,2} = \tau_2 u_2$.

Desired tension difference between the cables on the ground

The desired variable to control a linear combination of the two $F_{a,2} + \alpha F_{a,3} \rightarrow f^*$.

$$F_{a,1} = F_p - F_{a,2} - F_{a,3} \qquad \Longrightarrow \qquad \begin{aligned} F^*_{a,1} &= F_p - (f^* - \alpha F^*_{a,3} + F^*_{a,3}) \\ &= F_p - f^* - (1-\alpha)F_{a,3} \end{aligned} \tag{4.18}$$

For the specific case where $F_{a,2} - F_{a,3} \rightarrow f^*$:

$$F^*_{a,1} = F_p - f^* - 2F^*_{a,3} = F_p - f'^* \tag{4.19}$$

4.2 Extension to the Complete ACTS Architecture

At high level, the architecture that enables the system to reach the goal is shown in Figure 4.5.

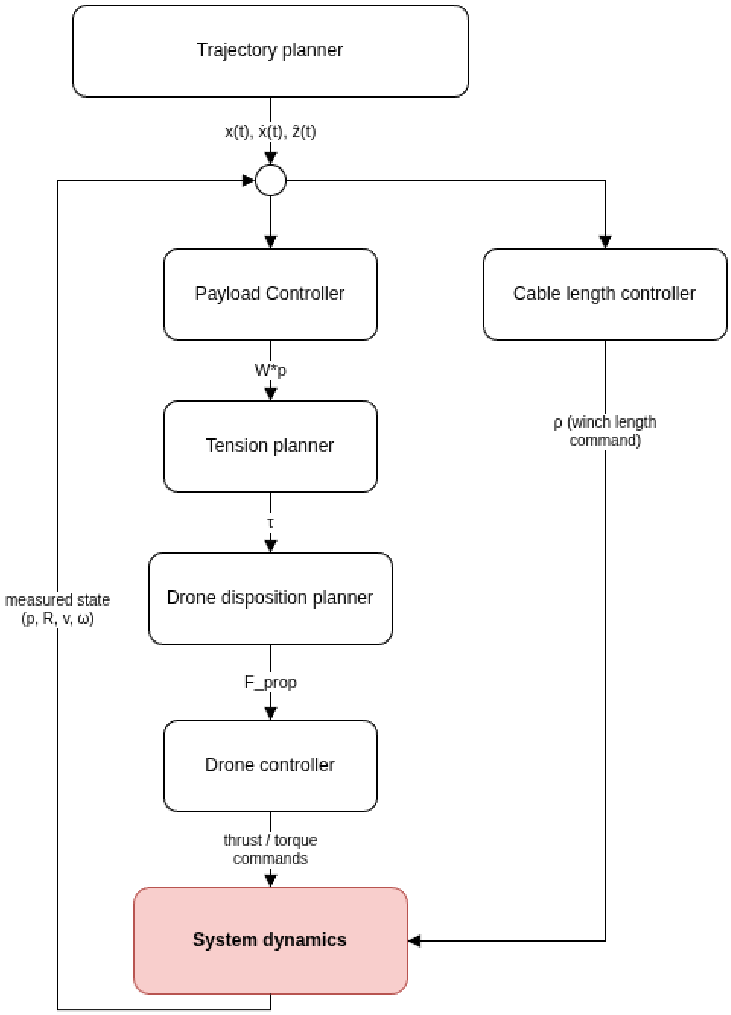


Figure 4.5: High level control scheme

Trajectory planner

The trajectory planner is able to compute the payload trajectory $x(t)$ and its velocity $\dot{x}(t)$ and acceleration $\ddot{x}(t)$ given the system's goal, such as:

1. Make the payload fly;
2. Bring the payload at the right height and orientation;
3. Approach the façade.

And/or the initial and final position of the payload.

Payload Controller

Then, the desired payload wrench W_p^* is computed for each point in the trajectory based on the previous state of the system. The desired payload wrench is computed from a geometric PD controller. The translational component is given by

$$F_p^* = m_p g + K_p(p^* - p) + K_d(\dot{p}^* - \dot{p}), \tag{4.20}$$

where p^* and p are the desired and current payload positions, respectively. For the rotational dynamics, the controller follows the geometric formulation on $SE(3)$ [20]. The attitude error and angular velocity error are defined as

$$\begin{aligned} e_R &= \frac{1}{2}\left(R^{*\top} R - R^\top R^*\right)^\vee \\ e_\Omega &= \Omega - R^\top R^* \Omega^* \end{aligned} \tag{4.21}$$

Where $(\cdot)^\vee$ denotes the inverse of the skew-symmetric (hat) operator. Since the controller performs set point regulation, the desired angular velocity is zero: $\Omega_d = \mathbf{0}$, yielding $e_\Omega = \Omega$. The desired control moment in the world frame is then

$$M_p^* = R(-K_R e_R - K_\Omega e_\Omega) \tag{4.22}$$

and the desired payload wrench becomes

$$W_p^* = \begin{bmatrix} F_p^* \\ M_p^* \end{bmatrix}. \tag{4.23}$$

Winch-Level Cable Length Controller

When transitioning the payload to a target pose along the planned trajectory, operational priority is given to the ground actuators due to their significantly higher mechanical stiffness and positioning precision compared to UAVs.

The ground subsystem is modeled as a 6-DOF Gough-Stewart parallel mechanism where the six ground cables act as active legs. Given the target payload position p^* and target orientation matrix R^*, the ideal length ρ_i^* for the i-th ground cable is defined geometrically by:

$$\rho_i^* = \|a_i - (p^* + R^* \, {}^p b_i)\| \tag{4.24}$$

To account for the finite velocity of the winch motors, the desired cable length is not applied instantaneously. Instead, the cable-length reference is updated incrementally at each control step according to

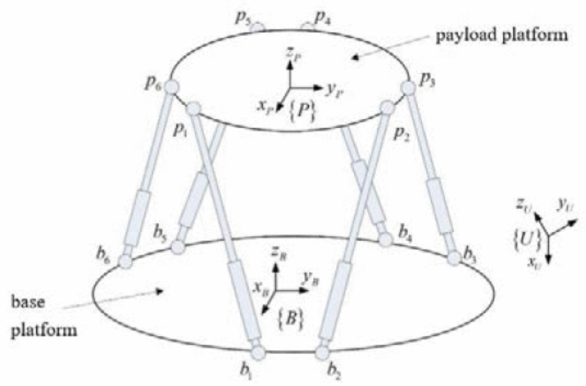


Figure 4.6: Gogh Stewart schematics [7]

$$\rho_i(t + \Delta t) = \rho_i(t) + \Delta\rho_i, \tag{4.25}$$

where the increment $\Delta\rho_i$ is limited such that $|\Delta\rho_i| = v_{\max}\Delta t$, with $v_{\max}$ being the maximum winch speed. The resulting cable-length reference is then sent to the actuator position controller. Consequently, the commanded cable length converges toward the desired value while respecting the physical velocity limits of the winch motors and avoiding unrealistically fast spool motions.

Tension planner

The tension planner is the layer responsible for guaranteeing that every cable stays taut throughout the motion. In fact, when working with ACTS, a payload wrench configuration is statically admissible if and only if there exists a set of strictly positive cable tensions ($\tau_i \geq \tau_{min} > 0$) that balance the target operational wrench. This is due to the fact that cables can only pull and not push. Mathematically, this layout optimization is formulated as:

$$\begin{aligned}
\min_{p_a} \quad & \sum_{i=1}^{n_a} \|F_{prop}(p_{a,i}, \tau_{optimal,i})\|^2 + K\|J_p\tau_{optimal} - W_p^*\|^2 \\
\text{subj. to} \quad & x_i^2 + y_i^2 + z_i^2 = l_i^2 \quad \forall i \in [1, n_a] \\
& \rho_{WCC}(\mathcal{U}) \leq 0 \\
& \rho_{IFC}(\mathcal{U}) \leq 0
\end{aligned} \tag{4.26}$$

Which is a Nonlinear optimization Problem (NLP) due to the Euclidean distance constraints.

This objective function aims to minimize the propulsion force of the drones while satisfying the wrench equilibrium equations on the payload. The parameter K acts as a penalty weight that prioritizes the tracking performance over energy efficiency; setting a high value (e.g., $K = 100$), heavily penalizes any mismatch between the generated wrench and the commanded target wrench W_p^*.

For whatever positions p_a, the cable tensions τ are determined by a second inner loop

$$\begin{aligned} \tau_{optimal} = \arg\min_{\tau} \frac{1}{2}\tau^\top H\tau \\ \text{subj. to} \quad J_p^\top(\mathbf{u})\tau = W_p^* \\ \tau_{min,i} \leq \tau_i \leq \tau_{max,i} \quad \forall i \in [1, n_a + n_g] \end{aligned} \tag{4.27}$$

The lower bound $\tau_{min,i} > 0$ in the wrench feasible condition is what enforces the taut-cable requirement.

As the winch-level control drives $\rho \to \rho^*$, tracking the payload's motion toward its target pose, and $\mathbf{u} \to \mathbf{u}^*$, the achievable wrench set used by the tension planner is itself changing over the trajectory, since it depends on $\mathbf{u}$. As a consequence, $\mathbf{W}_p^*$ may be reachable at the target configuration but momentarily unreachable at the actual configuration, with all tensions kept strictly positive. When this occurs, Equation 4.27 should relax the equality constraint $\mathbf{J}_p^\top(\mathbf{u})\,\tau = \mathbf{W}_p^*$ to a least-squares objective, while keeping $\tau_{min,i} > 0$ as a hard constraint.

$$\begin{aligned} \tau_{optimal} = \arg\min_{\tau} \left(\frac{1}{2}\tau^\top H\tau + \gamma\|J_p^\top(\mathbf{u})\tau - W_p^*\|^2\right) \\ \text{subj. to} \quad \tau_{min,i} \leq \tau_i \leq t_{max,i} \quad \forall i \in [1, n_a + n_g] \end{aligned} \tag{4.28}$$

This guarantees that the tension planner always returns a taut solution, one that approximates the desired wrench as closely as the actual geometry allows, rather than returning no solution at all during the transient.

For the initial stage of development, this general QP is replaced by a simplified strategy: every cable tension is fixed at the minimum admissible value,

$$\|\tau_i^*\| = \tau_{min}, \qquad \forall i \in [1, n_a + n_g]. \tag{4.29}$$

Drone disposition planner

Once the tension planner computes the scalar target tension magnitudes, the drones must dynamically adjust their orientation and propeller thrusts to impose these forces.

From the static equilibrium balance (Equation 3.13) the required propeller thrust vector is given by:

$$\mathbf{F}_{prop,i} = \mathbf{F}_{a,i} - \mathbf{F}_{g,i} = \mathbf{J}_{a,i}^\top\tau_i + m_i\mathbf{g} = T_i\,{}^0_i\mathbf{R}\,\mathbf{e}_3, \qquad T_i = \|\mathbf{F}_{prop,i}\| \tag{4.30}$$

which is then handed to the attitude controller, responsible for aligning the body z-axis with the commanded thrust vector.

Chapter 5

Simulation Environment and Experimental Setup

5.1 Role of Simulation in the Thesis

The development of the ACTS requires the interaction of several coupled mechanical and control components. A physics-based simulation environment is therefore used to evaluate the proposed architectures and control strategies before physical implementation.

Simulation provides a controlled environment in which the system can be tested under repeatable conditions. It also enables the evaluation of the controller without the risks and practical constraints associated with experiments on a physical system.

An additional advantage of using a physics engine is that the complete coupled equations of motion do not need to be explicitly derived and numerically integrated for each configuration. Instead, the mechanical bodies, joints, constraints, actuators, and contact properties are defined in the simulation model, while the physics engine performs the corresponding numerical integration. Nevertheless, the validity of the simulation remains dependent on the assumptions and simplifications made in the model.

5.2 Simulator Selection

Two simulation environments were investigated during the development of the thesis: Gazebo and MuJoCo. Gazebo was initially considered because of its integration with ROS2 and the availability of existing UAV models and visualization tools. However, difficulties were encountered when attempting to represent the closed cable-constrained structure of the ACTS. MuJoCo was therefore selected for the final simulation environment.

5.2.1 Gazebo Environment

Gazebo Harmonic [21] was initially selected as a potential simulation environment because of its established use in robotics and its integration with ROS2. However, the ACTS architecture presents a particular modeling challenge. Indeed, Gazebo assigns each physical element in the simulation a parent, generating a tree-like structure. While this does not constitute a problem for conventional articulated robot, ACTS contains ground cables constrained at both ends (resp. on the payload and on a point fixed on the ground), resulting in closed kinematic constraints.

Additionally, cables as soft body cannot be modelled in Gazebo, as it only manage rigid object. To go around the problem, they are modelled using multiple rigid links connected using universal joints. Although it solved the problem, implementing it resulted into an impracticable slow-down simulation.

Several preliminary simulations were nevertheless successfully implemented in Gazebo, including a pulley mechanism, a UAV connected to the ground through a cable, and a UAV connected to a payload. While individual components of the system could be simulated, such as pulley mechanism and single drone-cable simulation, extending this approach to the complete ACTS resulted in a more complex and computationally demanding model.

These experiments were useful for validating the initial modeling approach and identifying the main limitations before transitioning to another simulation environment.

Add photos

Given the additional modeling effort required to represent the complete ACTS architecture and the limited time available for the thesis, continuing the Gazebo implementation was not considered the most effective approach for obtaining the required simulation results.

5.2.2 MuJoCo Environment

MuJoCo (Multi-Joint dynamics with Contact) was selected as the final simulation environment because it is well suited to the closed-loop structure of cable-driven robots. In particular, MuJoCo provides tendon elements that can be used to represent cable-like connections between bodies, including constraints associated with their geometric configuration [22].

This capability significantly simplifies the representation of the ACTS compared with the rigid-link approximation investigated in Gazebo. The complete system can therefore be represented using the payload, UAVs, ground anchors, and cables while maintaining the closed-loop constraints required by the architecture.

For the simulations presented in this thesis, the cables are modeled as ideal geometric constraints. Their stiffness and damping are neglected, such that the cables transmit tensile forces without introducing additional elastic dynamics. This simplification is consistent with the quasi-static and geometric analysis used for the architecture evaluation, while allowing the simulation to focus on the influence of cable configuration, attachment geometry, and actuator placement.

5.3 Software architecture (UML)

All implementation code[1] is available on GitHub [23]. The repository is organized into three functional groupings:

- a `create_xml` package, an offline configuration and model-generation stage;
- an `acts_simulator` package, a runtime simulation stage;
- a `simulation_run` package, a mass-evaluation stage.

[1] `https://github.com/chiarabuono/acts_simulator.git`

The `create_xml` package handles architecture search and MuJoCo model generation offline, producing XML files that are loaded at startup.

A visual interface (`xml_generator.py`) is also provided to build a custom XML model interactively, ready to be tested without going through the automated architecture search.

The resulting models can be run by two independent codes: `acts.py`, a single-model, single-pose driver used for qualitative verification; and `batch_run.py`, a headless driver that iterates over every candidate XML model and every target pose in a batch pose list.

Both codes are meaningful for different reasons. `acts.py` runs one simulation at a time with a live visual interface; this makes it slower, but well suited to the qualitative verification of the controller, which is what it was used for in the early stages of development. Each run saves a single indices row to `collected_data`, together with a video recording and live plots of the tracking performance. `batch_run.py` executes the same underlying pipeline without the visual interface, which makes it considerably faster over many configurations: it accumulates one results row per (model, pose) combination directly into `results.xlsx`. This is the driver used to collect performance data across architectures and poses, from which the conclusions of Chapter 6 are drawn. This overall process is summarized in Figure 5.1.

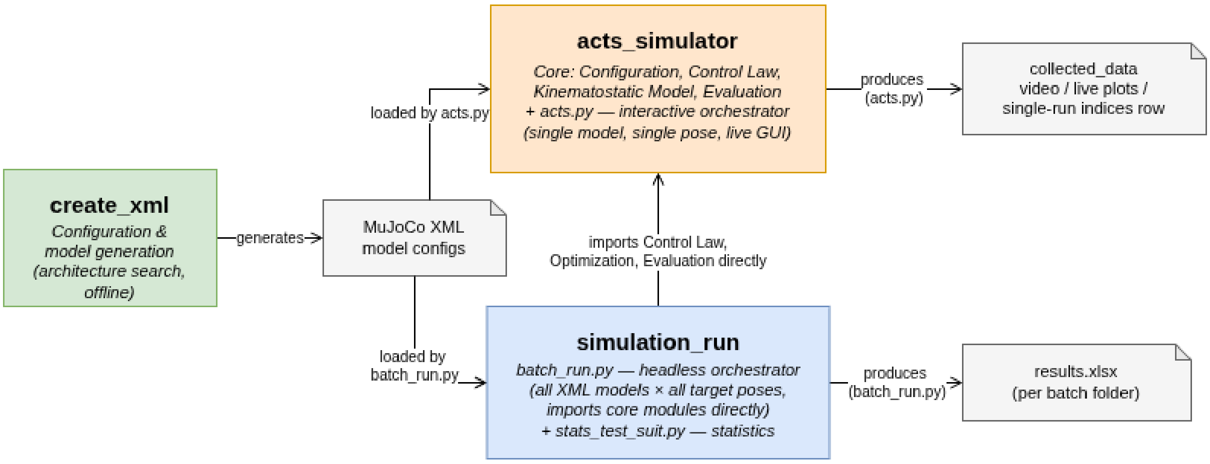


Figure 5.1: Architecture overview

Figure 5.2 details the internal organization of `acts_simulator`. `acts.py` runs the simulation, calling into three modules at each control step:

- the control-law layer (`utils_control.py`), which is described in Figure 5.3
- the kinematostatic/optimization layer (`utils_optimization.py`), which solves for the drone positions and cable tensions that best realize the desired payload wrench
- and the evaluation layer (`utils_performance_indices.py`), which compute the performance indices described in Section 3.3.2 and saves them to file.

The drone control-law layer introduced above is implemented as a small class hierarchy, as shown in Figure 5.3. In fact, a shared base class, `BaseDrone`, computes everything common to every configuration, such as the motor-mixing matrices, the attitude controller and the necessary rotor speeds. Each subclass replaces only `position_controller()`, providing the force law appropriate to what that drone is physically attached to.

The four subclasses correspond directly to the successive force-control cases developed in Chapter 4, and to the order in which the controller was actually developed and tested:

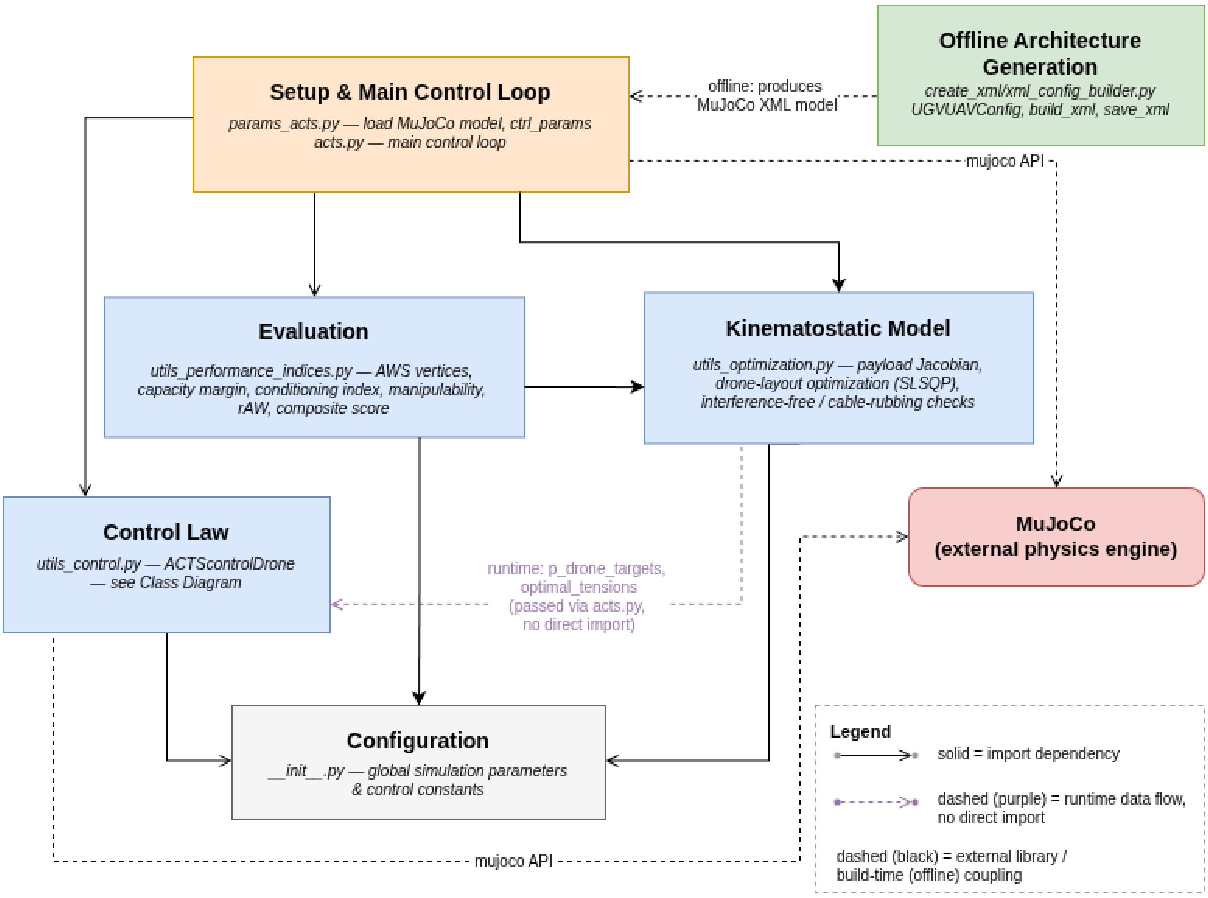


Figure 5.2: Package Diagram

- `FreeFlightDrone` implements the unconstrained case and was the first to be tested, to verify that a single drone could reach a desired position with no cable constraint at all;
- `CableTetheredDrone` extends this to a drone tethered directly to a fixed ground anchor by a cable described in Section 4.1.1
- `PayloadControlDrone` extends the cable case further to include a payload described in Section 4.1.2;
- `ACTScontrolDrone` is the controller ultimately used in the full ACTS architecture, driven directly by `acts.py`.

This progressive structure allowed each force-control case to be validated in isolation before being combined into the final system, and it keeps `position_controller()` as the single, well-defined extension point where a new case could be added without touching the shared attitude and motor-mixing logic.

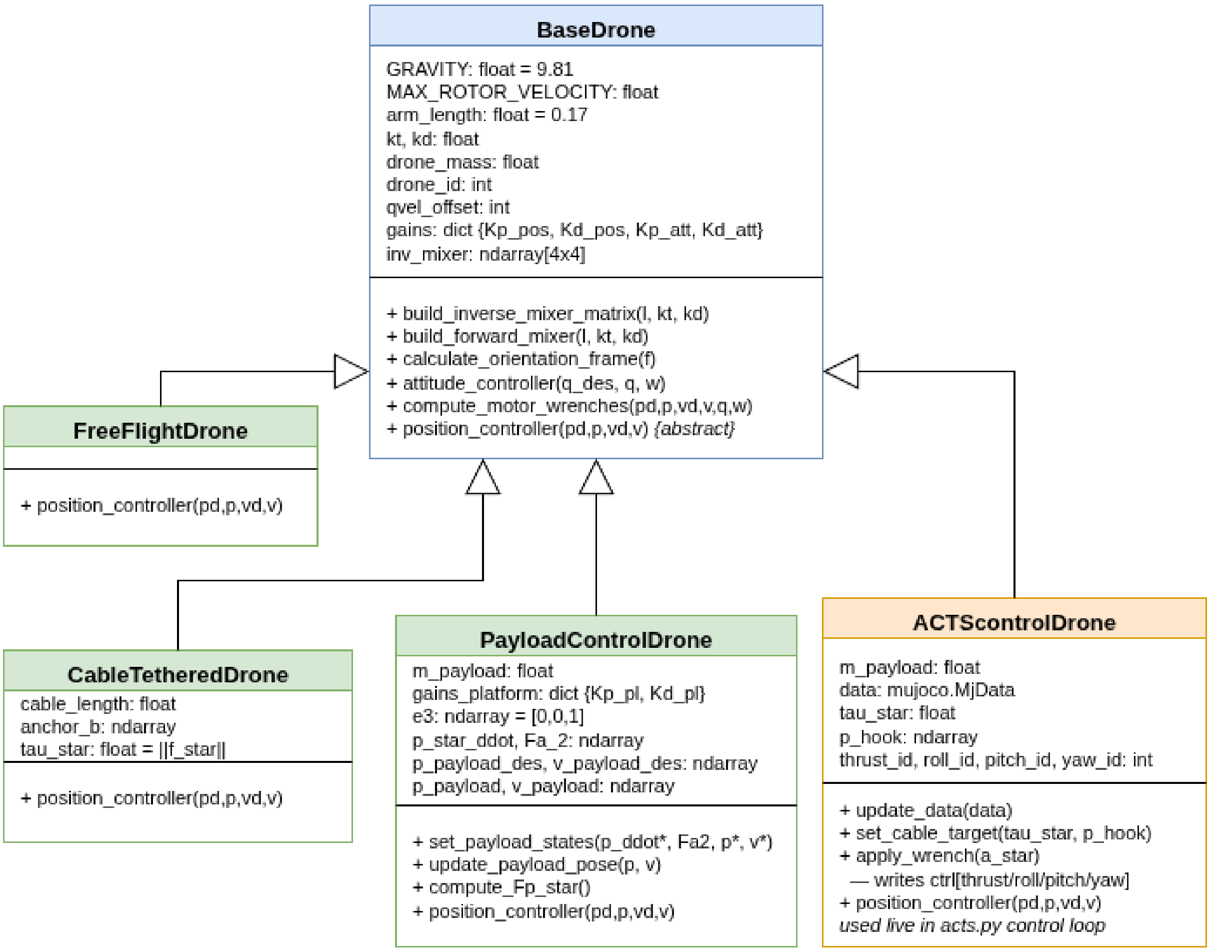


Figure 5.3: Drone Controller Class Diagram

5.4 Simulation setup

The simulations use the same payload, UAV model, cable properties, controller parameters, and numerical settings to ensure a fair comparison between different architectural configurations. These values are shown in Table 5.1. Additionally, the simulation time is equal to 0.002 seconds.

The architectural variables investigated in Chapter 6 are the ground-cable routing, ground-anchor placement, and UAV attachment geometry. At the start of the simulation, the payload is on the ground along with the drones, which are resting on its top surface.

5.5 Simulation methodology and evaluation framework

The geometric and static performance indices provided during the model generation stage, describe the instantaneous ability of a given architecture to generate payload wrenches and identify configurations that are close to kinematic singularities. However, these indices are evaluated under static or quasi-static assumptions and therefore cannot fully characterize the behavior of the closed-loop system during motion. In the latter case, cable directions, tensions and payload dynamics continuously evolve. Consequently, a

Simulation parameter	Value
Payload and initial conditions	
Payload mass	1 kg
Initial payload position	$[0.0, 0.0, 0.0]^\top$ m
Initial payload orientation	$\mathbb{I}$
Cable and winch parameters	
Minimum cable tension τ_{min}	5.5 N
Maximum cable tension τ_{max}	150 N
Maximum winch speed $v_{\mathrm{winch,max}}$	1.5 m/s
UAV actuation parameters	
Maximum rotor velocity ω_{max}	1666 rad/s
Thrust coefficient k_t	5.5×10^{-6}
Drag coefficient k_d	3.299×10^{-7}
Minimum thrust T_{min}	1.0 N
Maximum thrust T_{max}	$4k_t\omega_{\mathrm{max}}^2$
Safety constraints	
Minimum UAV–UAV distance	0.4 m
Minimum cable–cable distance	0.05 m
Evaluation parameters	
Position tolerance	0.1 m
Orientation tolerance	3°

Table 5.1: Fixed parameters used in the numerical experiments.

configuration that appears favorable according to static indices may still exhibit oscillations, poor tracking or instability. Closed-loop simulation is therefore necessary to verify whether the theoretical advantages translate into practical performance.

The resulting analysis distinguishes between architectures that are favorable according to the analytical performance indices and those that provide favorable closed-loop behavior in simulation.

Chapter 6

Simulation results

The objective of this chapter is to evaluate how the architectural design of an ACTS influences its ability to accurately and robustly manipulate a payload. The focus is on how different cable routing strategies, UAV attachment configurations, and ground hook placements influence the overall system performance.

In fact, a configuration is considered successful if the desired payload is reached with sufficiently small position and orientation errors, all cables remain under positive tension throughout the maneuver, and the payload converges to an equilibrium without persistent oscillations.

First, the classical Stewart configuration is analyzed and used as a reference architecture due to its geometric symmetry and extensive use in the literature. In fact, if even Stewart-like performs poorly, the problem is likely the controller rather than the architecture.

Two desired payload poses are taken in considerations. In the first scenario, the desired payload orientation coincides with the world reference frame; while in the other scenario a non-zero desired orientation is imposed. This allows to analyze separately the rotational motions from the translation tracking.

The remaining experiments then investigate how modifications to the ground cable routing, UAV attachment geometry, and hook placement affect both the theoretical performance indices and the closed-loop behavior of the system. This progressive analysis makes it possible to identify which geometric features have the greatest influence on payload manipulation performance.

6.1 Stewart platform baseline

The Stewart platform configuration is used as a baseline for evaluating the proposed ACTS architectures. Its symmetric arrangement of the six ground cables provides a well-defined reference configuration against which alternative cable routings can be compared. The baseline is first examined using the static performance indices introduced in Section 3.3.2.

In particular, the conditioning index, manipulability, radius of available wrench, capacity margin, and composite score are evaluated at the prescribed payload pose. These quantities provide a reference for assessing the wrench-generation capability and isotropy of the ground-cable arrangement. The complete ACTS configuration is then evaluated by incorporating the aerial cables and their actuation constraints. Finally, the resulting configuration is simulated in MuJoCo to verify the dynamic payload tracking behavior.

6.1.1 Numerical Verification of the Baseline Configuration

This section verifies that the implemented optimization and control algorithms correctly reproduce the mathematical model derived in Chapters 3. For one representative Stewart-platform configuration, each stage of the computation is independently reconstructed from the governing equations: cable directions are computed from the geometry, the resulting Jacobian is assembled, cable tensions are substituted to recover the payload wrench, and the required UAV thrust is recomputed from the static equilibrium equations. Agreement between these hand calculations and the implementation provides confidence that subsequent comparisons between ACTS architectures reflect architectural differences rather than implementation errors.

Configuration

Payload dimensions: $2.0 \times 2.0 \times 0.2$ m
Desired pose: $p^* = [0.5, -0.5, 2.0]^\top$ m
Desired orientation: $R^* = \mathbb{I}$
Desired payload wrench: $W_p^* = [0, 0, 9.81, 0, 0, 0]^\top$

attachment		hook points	
$A_4 \equiv A_5$	$[0.0, 1.1, 0.0]^\top$	$B_4 \equiv B_9$	$[0.5, 0.05, 1.9]^\top$
$A_6 \equiv A_7$	$[-0.9526, -0.55, 0.0]^\top$	$B_5 \equiv B_6$	$[-0.0237, -0.775, 1.9]^\top$
$A_8 \equiv A_9$	$[0.9526, -0.55, 0.0]^\top$	$B_7 \equiv B_8$	$[0.9763, -0.775, 1.9]^\top$

Table 6.1: Ground anchor and hook attachment points for the Stewart baseline.

Using Equation 3.6b, the ground cable directions u_i are:

$$\begin{aligned}
u_4 &= [-0.2245,\ 0.4713,\ -0.8529]^\top & u_7 &= [-0.7100,\ 0.0828,\ -0.6993]^\top \\
u_5 &= [-0.0089,\ 0.7024,\ -0.7117]^\top & u_8 &= [-0.0124,\ 0.1176,\ -0.9930]^\top \\
u_6 &= [-0.4545,\ 0.1047,\ -0.8846]^\top & u_9 &= [0.2215,\ -0.2937,\ -0.9299]^\top
\end{aligned}$$

Table 6.2: Ground cable unit directions.

$$J_{p,\text{ground}} = \begin{bmatrix}
-0.22445 & 0.47135 & -0.85291 & -0.42197 & 0.02245 & 0.12345 \\
-0.00888 & 0.70238 & -0.71175 & 0.26597 & -0.33812 & 0.33656 \\
-0.45452 & 0.10475 & -0.88456 & 0.25373 & -0.37586 & 0.29632 \\
-0.70998 & 0.08282 & -0.69934 & 0.20060 & 0.40409 & -0.52834 \\
-0.01239 & 0.11759 & -0.99298 & 0.28483 & 0.47420 & -0.47636 \\
0.22151 & -0.29365 & -0.92989 & -0.54081 & -0.02215 & -0.12183
\end{bmatrix} \tag{6.1}$$

Drone Attachment Geometry

Unlike the ground anchors, the drone attachment points a_1, a_2, a_3 are not fixed geometry — they are the output of the layout optimization of Section 4.2. The corresponding cable directions u_1, u_2, u_3 are therefore extracted from a converged simulation run for this configuration rather than computed from known attachment coordinates:

$$a_1 = [1.1020, -0.2692, 3.5972]^\top, \quad a_2 = [0.1802, -0.7553, 3.4976]^\top, \quad a_3 = [0.9397, -1.2760, 3.5192]^\top. \tag{6.2}$$

With $r_i = p^* + R^* \, {}^p b_i$ and the drone hook offsets ${}^p b_i$, the resulting drone cable directions are

$$u_1 = [0.0837, -0.0294, 0.9961]^\top, \quad u_2 = [0.1042, -0.3528, 0.9299]^\top, \quad u_3 = [0.2926, -0.1504, 0.9443]^\top, \tag{6.3}$$

with $\|a_i - r_i\| \approx 1.503$ m for all three, consistent with the nominal drone cable length of 1.5 m and confirming the optimizer converged to a geometrically taut solution. The corresponding drone Jacobian is

$$J_{p,\text{drones}} = \begin{bmatrix} 0.0837 & 0.1042 & 0.2926 \\ -0.0294 & -0.3528 & -0.1504 \\ 0.9961 & 0.9299 & 0.9443 \\ 0.2769 & 0.2910 & -0.5043 \\ -0.4661 & 0.4533 & 0.0293 \\ -0.0370 & 0.1394 & 0.1609 \end{bmatrix}. \tag{6.4}$$

Cable Tensions

For this configuration, the optimizer returns

$$\tau_{\text{UAV}} = [13.6970,\ 11.1548,\ 14.4154]^\top \text{ N}, \qquad \tau_{\text{ground}} = [5.4903,\ 5.4740,\ 5.4901,\ 5.4784,\ 5.4860,\ 5.4708]^\top \tag{6.5}$$

All six ground tensions lie close to $\tau_{\min} = 5.5$ N. This behavior is physically expected. Because the desired wrench is almost entirely vertical, the UAVs provide most of the lifting force while the Stewart cables primarily stabilize the payload orientation and prevent lateral motion. Consequently, the optimizer drives the ground tensions toward their minimum admissible values, maintaining taut cables while minimizing unnecessary internal forces.

Payload Wrench Verification

Combining $J_{p,\text{drones}}$ and $J_{p,\text{ground}}$ into the full Jacobian J_p and evaluating $W_p = J_p \tau$ with the tensions above:

$$W_p = J_p \tau = \begin{bmatrix} -0.0023 \\ 0.0018 \\ 9.8110 \\ -0.0003 \\ -0.0001 \\ -0.0009 \end{bmatrix}, \qquad W_p^* = \begin{bmatrix} -0.0158 \\ 0.0180 \\ 9.7341 \\ -0.0022 \\ 0.0019 \\ -0.0151 \end{bmatrix}. \tag{6.6}$$

All six components match to within 0.08 in force units and 0.015 in moment units. This residual is expected rather than an error: with `MODE = "tau_optimal"`, tensions are found by a bounded least-squares solve (Eq. 4.27) that *penalizes* wrench mismatch rather than enforcing $J_p^\top \tau = W_p^*$ as a hard equality constraint, so a small, non-zero residual is the expected outcome of the optimization rather than a sign of implementation error.

Required Drone Thrust

Under the steady-state assumption $p_d = p$, $\dot{p}_d = \dot{p} = 0$, Eq. 4.30 reduces to $F_{prop,i} = \tau_i u_i + m_i g \, e_3$, giving the hand-computed thrust magnitudes

$$T_1 = 33.29 \text{ N}, \qquad T_2 = 30.27 \text{ N}, \qquad T_3 = 33.57 \text{ N}. \tag{6.7}$$

The controller-computed thrust commands for the same converged pose are 33.2793, 30.2683, and 33.5657 N respectively, matching the hand-computed values to within floating-point precision and confirming that `compute_motor_wrenches` correctly implements Eq. 4.30.

Discussion

Both checks confirm the implementation is correct, at two different tolerances appropriate to what each is verifying. The thrust-command comparison, which involves no optimization step, matches to floating-point precision, confirming that the geometric force-to-thrust mapping in the drone controller (Eq. 4.30) is implemented exactly as derived. The wrench-tracking comparison shows a small, non-zero residual, which is the expected signature of the tension planner's least-squares formulation (Eq. 4.27) rather than an error: the optimizer trades a small wrench-tracking residual for reduced drone propulsion effort, weighted by K in the objective of Eq. 4.27. Together, these results validate both the geometric optimization and the low-level control implementation for this representative configuration, providing the basis for the comparative architecture study that follows.

6.1.2 Comparison between tension allocation strategies

As mentioned in Section 4.2, we evaluate two distinct tension allocation strategies:

- Minimal tension strategy, in which a minimum cable tension is imposed to each cable tension such that $\|\tau_i\| = \tau_{min}$.
- Optimal tension strategy, in which cable tensions are determined by the quadratic optimization problem in Equation 4.28

The performance indices and tracking errors under both strategies across two operational scenarios are summarized in Table 6.3.

The minimal tension strategy has negative consequences on the system's robustness. By locking all cable tensions to their absolute lowest limit ($\tau_{\min}$), the range of available tension choices shrinks to a single point. As a result, the controller loses its ability to adjust forces, reducing the payload's Available Wrench Set (AWS) to just a single fixed outcome. [24] As a result, any deviation of the desired payload wrench W_p^* from this single point results in a negative capacity margin (as seen in Scenario A where $\gamma = -0.19343$). This indicates that the system is theoretically incapable of resisting external disturbances or maintaining equilibrium without violating the tension constraints.

Switching to the optimal tension allocation, allows the cable tensions to vary dynamically between τ_{min} and τ_{max}. As a result, the radius of the Available Wrench increases and the capacity margin takes on a positive value ($\gamma = 10.46403$). Thus, also the composite score increases. Differently, changing the tension allocation algorithm has no effect on the conditioning index or manipulability because they are kinematic metrics that only depend on the platform's geometric position.

On the other hand, it is worth to notice that, since the objective function does not contain any state-tracking terms, this optimizer does not directly minimize the payload's real-time position or orientation tracking errors. Because of this, a spatial configuration that yields globally minimal propulsion effort does not guarantee a minimum in tracking error.

Metric / Parameter	Scenario A		Scenario B	
	τ_{min}	τ_{opt}	τ_{min}	τ_{opt}
Desired Position	$[0.5, -0.5, 2.0]^\top$		$[0.5, -0.5, 2.0]^\top$	
Desired Orientation	$[1.0, 0.0, 0.0, 0.0]^\top$		$[1.0, 0.1, -0.1, 0.1]^\top$	
Conditioning Index	0.10259	0.09617	0.07052	0.05550
Manipulability	0.47775	0.43700	0.31471	0.24108
Radius Avail. Wrench	49.29711	47.90399	52.21278	53.22707
Capacity Margin	-0.19343	10.46403	-3.83256	2.65386
Worst-case CM	0.90998	0.76401	0.90998	0.75072
Composite Score	17.35180	20.62849	17.64446	19.42488
Position Error	0.00201	0.00284	0.00187	0.00262
Orientation Error	0.00121	0.00209	0.00379	0.00385

Table 6.3: Comparative performance indices and tracking errors for Stewart baseline configuration across Scenarios A and B.

Although the Stewart configuration exhibits good overall performance and provides an excellent reference architecture, the proposed ACTS differs from a classical Stewart platform because the UAV cables can be connected to the payload in several different ways. Moreover, different ground cable routings may further improve specific performance aspects such as stability, wrench generation capability, or robustness. For these reasons, the Stewart configuration is adopted as a baseline against which all alternative architectures are compared.

An ideal ACTS architecture should remain far from singular configurations, generate payload wrenches efficiently in every direction, and maintain sufficient redundancy to accommodate disturbances while respecting cable tension limits. Consequently, all the considered performance indices should be as big as possible.

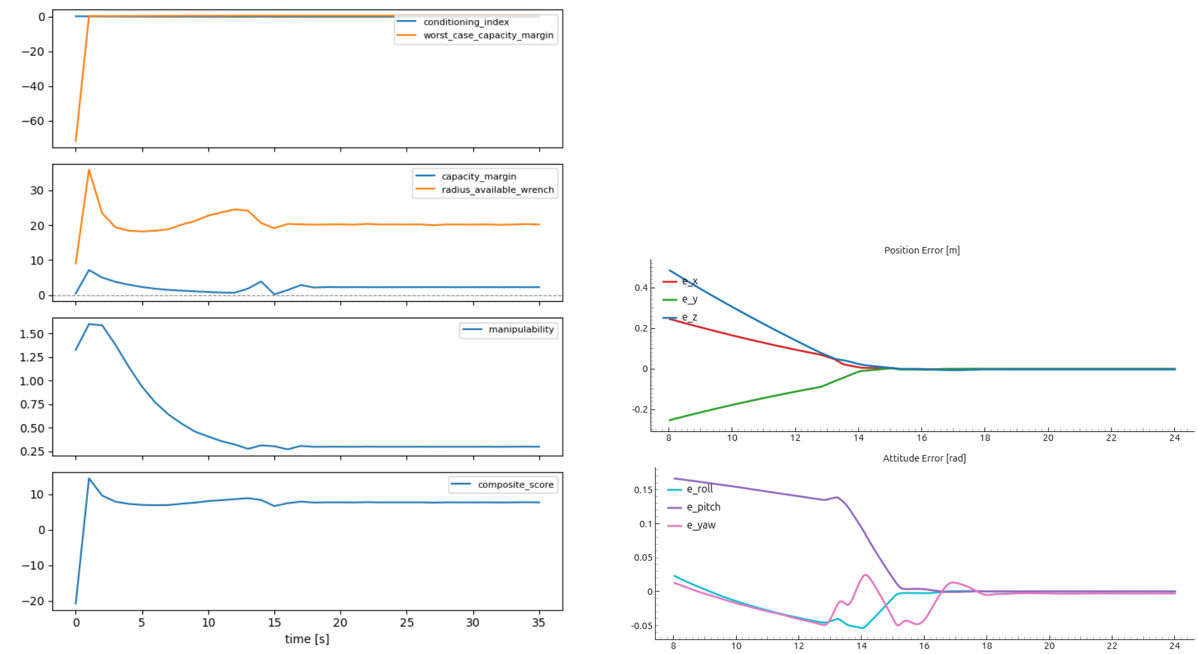
The considerations from now on will be done using the optimal tension strategy.

6.2 Ground cable routing

We have already talken about it in the previous sections, how to do it?

Once the controller has been validated using the Stewart baseline, we can proceed to consider different ways to connect six ground cables to the platform (Figure 6.2) and the drone (Figure 6.3). The same ground configuration hook-anchor can change the system's stability based on which hook is connected to which anchor point on the ground. Therefore, we will start considering the drone attachment configuration as fixed to the triangular arrangement and vary only the routing of the six ground cables. This specific UAV configuration is chosen because it provides a symmetric distribution of the three drones around the payload. This approach allows one to isolate the influence of the ground cable routing and attribute differences in performance solely to the ground architecture. Finally, in Section 6.4, the impact of various UAV attachment geometries is examined independently.

Focusing on the ground cable routing, there are at most $6! = 720$ physically distinct valid ways to route the six ground cables. Scoring an architecture using only one arbi-



(a) Performance indices

(b) Pose error

Figure 6.1: Graphs

trarily chosen routing raise the question on how lucky the routing choice was and how good the architecture actually is.

To eliminate this source of bias, an exhaustive routing search is performed for every pair of payload and ground layouts. All physically valid routings are enumerated while respecting the maximum cable capacity of each attachment point. For every routing, the corresponding payload Jacobian J_p is constructed according to Equation 3.7, and the routing is evaluated over a representative set of payload poses. At each pose, the routing must satisfy the necessary conditions introduced in Section 3.3.2. Routings that fail any of these conditions are discarded. Among the surviving candidates, the routing is assigned its worst-case performance across all poses, and the architecture is finally characterized by the routing that maximizes the conditioning index. In this way, architectures are compared according to their best achievable routing rather than to an arbitrary cable assignment. Architectures are subsequently ranked against one another by worst-case conditioning index.

Update with correct data

Pay. Layout	**Gnd. Layout**	N_{rout}	N_{pos}	**Feas.**	ν	w	$\sigma_{\min}$	$\sigma_{\max}$	**Rank**	**Best Routing**
2X3	pentagon	360	8	✓	0.1402	0.3680	0.3275	2.3452	✓	A-E—B-A—C-D—D-C—E-B—F-E
two-triangles	pentagon	360	8	✓	0.1383	0.2169	0.3241	2.3566	✓	A-E—B-A—C-C—D-D—E-B—F-E
pentagon	2X3	360	8	✓	0.1354	0.3773	0.3169	2.3401	✓	A-A—B-F—C-E—D-B—E-C—E-D
hexagon	pyramid	720	8	✓	0.1343	0.1664	0.3180	2.3794	✓	A-D—B-A—C-B—D-E—E-F—F-C
pentagon	pentagon	168	8	✓	0.1316	0.2494	0.2899	2.3420	✓	A-E—B-C—C-B—D-E—E-A—E-D

Table 6.4: Ground Cable Routing and Layout Performance Analysis

The composite score N is not used for ranking routings during the exhaustive search because it requires information that is only available after solving the tension planner, such as the optimal cable tensions and the available wrench set. Computing its capacity-

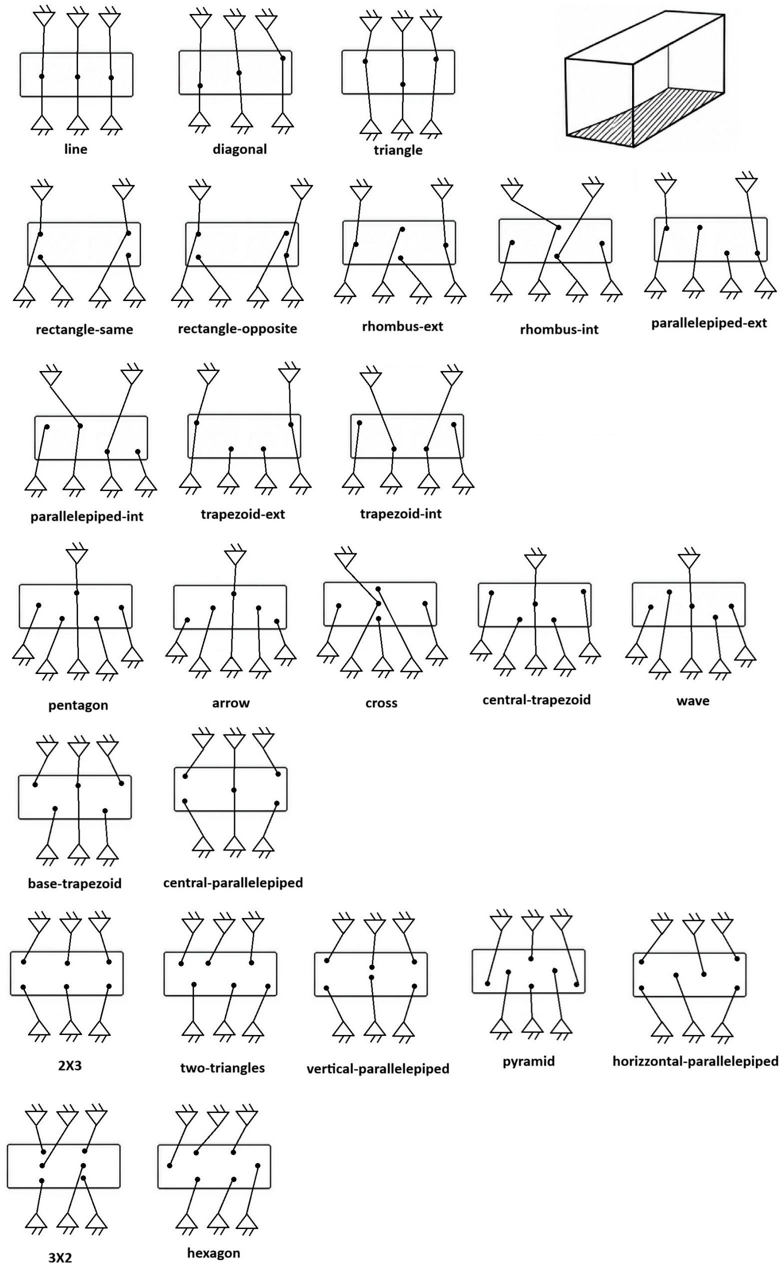


Figure 6.2: Possible 6 UGVs connections

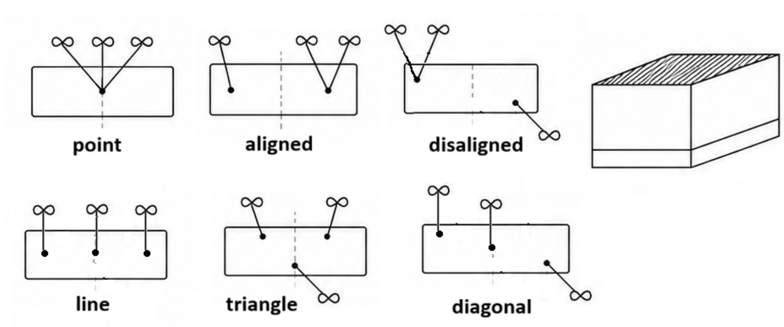


Figure 6.3: Possible 3 UAVs connection with one cable

based terms is also computationally expensive and unreliable near singular configurations. Additionally, its weighting factors have not been experimentally validated, making it unsuitable for selecting the best routing. Instead, the search uses the conditioning index (with manipulability as a supporting metric), as it is a parameter-free, geometry-based measure that can be efficiently computed directly from the Jacobian. The composite score and capacity-based metrics are therefore evaluated later during closed-loop simulations, once a routing and tension solution are available.

Algorithm 1 Cable Routing Optimization and Architecture Ranking

for all (payload_layout, ground_layout) pairs **do**
    Enumerate every valid 6-cable routing
    for all routing **do**
        for all pose $\in$ test-pose set **do**
            Build J_p ▷ Eq. 3.7
            Check Wrench Closure ▷ $\text{rank}(J_p) = 6$
            Check Wrench Feasibility ▷ $0 \leq \tau_{\text{min}} \leq \tau \leq \tau_{\text{max}}$
            Check Interference-Freedom ▷ Eq. 3.19
        end for
        Record this routing's **worst-case** score across all poses
    end for
    Keep the **best** surviving routing for this architecture
end for

How can I see if a configuration is more inclined to singularity only by looking at it?

Line or diagonal results in not-feasible configurations, why? What does this mean? Geometric reasons?

The line and diagonal configurations approach kinematic or wrench singularities, where the cable arrangement cannot generate the desired six-dimensional wrench while maintaining positive tensions. Conversely, the Rectangle-Same configuration remains kinematically feasible but exhibits poor rotational stiffness due to its asymmetric cable pairing. As a result, although the Jacobian remains full rank, the payload develops a rocking motion that cannot be predicted by the geometric performance indices alone.

The theoretical indices evaluate the geometric and static properties of a cable routing. They identify configurations that are close to kinematic or wrench singularities and rank architectures according to their ability to generate payload wrenches. However, they cannot predict the transient response of the closed-loop system. Dynamic effects such as payload inertia, controller gains and load redistribution are only captured by simulation. Consequently, while poor theoretical indices almost always indicate poor dynamic performance, the opposite is not necessarily true: a routing with good static indices may still exhibit oscillations or poor tracking.

Insert table comparing the two

Rectangle-same really unstable configuration

The Jacobian is full rank and the static indices are acceptable, yet the payload rocks because of an uneven load distribution and poor torsional stiffness. That example provides strong evidence that the theoretical indices are necessary but not sufficient for predicting closed-loop behavior.

Talk about triangle-triangle combination

Other combinations that give interesting results?

6.3 Ground anchor placement

Increasing the spacing between the ground hooks enlarges the available wrench set and moves the system farther from wrench singularities, explaining the observed improvement in tracking performance.

6.4 Drone hook placement

How to structure this chapter:

1. Why did we choose to go for the method: controller and see if that pose is reachable? [DONE]
2. Talk about Stewart configurations,
   - which are the indices values [DONE]
   - what these can tell us about everything [DONE]
   - How do I want each performance index? [DONE]
   - Show them in a graph
3. Given an UAV configuration (we chose triangle), analyze cable routing
   - Why we choose the triangle for the UAVs configuration? [DONE]
   - How the code work, how a given cable routing has been found [DONE]
   - How the results before simulation and after simulation differ?
   - Why we need to test it dynamically and not just trust the results we obtain before? [DONE, add table]
   - Rectangle-same really unstable configuration
   - line or diagonal results in not-feasible configurations, why? What does this mean? Add geometric reasons
4. Given the results of the previous step, let's analyze the first 3/5 configurations with all 6 possible UAV configurations
   - Expectation: triangle being the best one, if not, which one is best? Rerun step 2 and see how things change
   - How drone attachment influences reaching the goal? Show evolution graph
5. Can a more spread-out hook configuration on the ground better the performance, or it is irrelevant?
   - Does the answer change when the desired position is If so, it opens results to the moving hook point. Further away?

- Show example on the top configuration and a terrible one.
- Are there any papers/theorem that says so?
- Show evolution graph

6. How a configuration far away from the hook point on the ground performs?
   (a) What Stewart does and what top configuration does
   (b) The hook position should cover more space on the ground. Does this better the result Stewart vs top configuration?
   (c) Is there a geometric reason/ theory for that?

Chapter 7

Conclusions

Future developments

- Consider different configuration rather then 6 on the ground and 3 on the top
- Façade interaction

Also answering these questions:

- Is it better having the drone connected through one or two cables to the payload in terms of easy maneuverability, feasible workspace, available wrench set and energy consumption?
- How accuracy on drones' position affects the accuracy on the platform's position?
- Are UAVs (resp. UGVs) strictly connected to the upper (resp. down) face or can they be attached to the down (resp. upper) face?

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Appendix A

Singularities and Performance Indices

write it better

Cable-driven parallel robots (CDPRs) and aerial cable-driven systems experience critical operating limits where force transmission or controllability degrades. Table A.1 summarizes these operational singularities and their physical consequences.

Type	**Cause**	**Detection Index**	**Consequence**
Kinematic	$\mathrm{rank}(J_p) < 6$	Condition Number, Manipulability	Loss of platform Degrees of Freedom (DoF)
Wrench	Desired wrench outside AWS	Capacity Margin, RAW	Target wrench cannot be generated
Tension	Negative or zero cable tensions	Tension optimization	Cable slack / loss of constraint
Stiffness	Poor cable distribution	Stiffness Matrix, Simulation	Unwanted oscillations and rocking
Dynamic	Closed-loop instability	Eigenvalues, Simulation	Trajectory tracking failure

Table A.1: Classification of Singularity Types in Cable-Driven Parallel Systems